A Siamese Deep Learning Framework for Bi-Temporal Hyperspectral Image Change Detection

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By

SAI KUMAR KOLLI - 228W1A12A0

JAYA PRAVEEN GALI - 228W1A1286

Under the Guidance of

Dr. Vaddi Radhesyam,

Associate Professor

DEPARTMENT OF INFORMATION TECHNOLOGY

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(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this project report titled **"A SIAMESE DEEP LEARNING FRAMEWORK FOR BI-TEMPORAL HYPERSPECTRAL IMAGE CHANGE DETECTION"** is a Bonafide record of work done by **SAI KUMAR KOLLI 228W1A12A0 and JAYA PRAVEEN GALI 228W1A1286** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

Dr. Vaddi Radhesyam	**(Dr. M. Suneetha)**
Associate Professor	Dean R&D, Professor &Head
Dept. of Information Technology	Dept. of Information Technology

EXTERNAL EXAMINER SIGNATURE

Date of examination

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R. SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	Project Title	A SIAMESE DEEP LEARNING FRAMEWORK FOR BI-TEMPORAL HYPERSPECTRAL IMAGE CHANGE DETECTION
2	Student Names & Numbers	K. Sai Kumar (228W1A12A0) G. Jaya Praveen (228W1A1286)
3	Name of The Guide	Dr. Vaddi Radhesyam
4	Research Group	Hyperspectral Image Processing and Remote Sensing
5	Application Area	Remote Sensing, Hyperspectral Image Analysis, Change Detection
6	Aim of the Project	To design a Siamese deep learning framework for accurate change detection in bi-temporal hyperspectral images by learning spectral-spatial features and generating reliable change maps.
7	Project Outcomes	1.Automated identification of changed and unchanged regions. 2.Accurate generation of change probability and binary change maps. 3.Improved efficiency for large-scale hyperspectral change analysis.

Student Signatures

1.

2.

Signature of the Guide:

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ABSTRACT

Accurate change detection in bi-temporal hyperspectral images has significant applications in areas like land use monitoring, urban growth analysis, agriculture evaluation, and environmental monitoring. However, detecting changes in hyperspectral images is a difficult task because of the high dimensional spectra, the presence of noise, and complex scenes. In this paper, we propose a Siamese deep learning architecture for change detection in bi-temporal hyperspectral images. A pre-processing step based on the Change Vector Analysis (CVA) approach is adopted to highlight spectral changes between images collected at different times, which improves the extraction of change-related information. The designed network can learn spectral-spatial characteristics in bi-temporal hyperspectral images and produce pixel-level change probability maps and binary change maps for the precise identification of changes. Quantitative metrics used for assessing the performance of the proposed model include overall accuracy, precision, recall, F1 score, Dice coefficient, Intersection over Union (IoU), and Area Under the Curve (AUC). Moreover, qualitative evaluations such as comparing the spectral signatures, performing correlation analysis, and evaluating the probability distribution are conducted to ensure the effectiveness of the proposed architecture.

Keywords: Bi-temporal hyperspectral image change detection, Siamese deep learning, Change Vector Analysis (CVA), hyperspectral imaging, remote sensing, spectral-spatial analysis.

CHAPTER-1

INTRODUCTION

This chapter covers the introduction of the project background and motivations, necessary concepts for the understanding of the study, problem statement, significance of the problem in reality, and the approach to be used in the research process.

1.1 ORIGIN OF THE PROBLEM:

Recently, the field of hyperspectral remote sensing has gained importance in detecting any changes occurring in the Earth's surface features. Bi-temporal hyperspectral images are extremely helpful due to their high spectral resolution and numerous bands, and hence, have found their applications in land-use change detection, urban expansion monitoring, agriculture assessment, environmental monitoring, disaster management, and so on. Hyperspectral imagery has several advantages over traditional remote sensing images in terms of detecting spectral differences which cannot be detected using traditional imagery. Nevertheless, hyperspectral image change detection is still a complicated problem due to its multi-spectral property.

The conventional methods used to perform change detection on hyperspectral data are either based on visual inspection or are classical image processing algorithms. The former is extremely tedious and impractical for large hyperspectral scenes whereas the latter includes algorithms like change vector analysis (CVA), spectral angle mapper (SAM), and iteratively reweighted multivariate alteration detection (IR-MAD). All these algorithms have several drawbacks which limit their utility in performing change detection tasks on hyperspectral images.

As deep learning technology continues to grow rapidly, various advanced techniques have been developed for hyperspectral imaging analysis tasks. It is possible for the deep learning algorithms to learn meaningful spectral and spatial features from data automatically and eliminate the requirement for manually extracted features. Siamese deep learning models can detect and analyze changes between two images captured at two separate times successfully by learning associations among them. Therefore, it

becomes easier for Siamese deep learning models to learn the relationships and differences between the images effectively.

In this research, a Siamese deep learning model is employed to detect changes in bi-temporal hyperspectral images. To preprocess the data prior to change detection process, CVA-based pre-processing will be conducted to emphasize the differences between pixels in two images and facilitate the change classification task. A Siamese deep learning framework will generate pixel-level and binary change maps for each of the two test datasets. Both quantitative measures like accuracy, precision, recall, F1-score, dice coefficient, intersection-over-union, and AUC score and qualitative measures such as spectral signature analysis and correlation analysis will be utilized to evaluate the performances of our proposed system.

1.2 BASIC DEFINITIONS AND BACKGROUND:

1.2.1 Hyperspectral Imaging

Hyperspectral imaging is a remote sensing technique that captures image data across hundreds of narrow and contiguous spectral bands. Unlike conventional RGB images, hyperspectral images provide detailed spectral information for each pixel, which helps in identifying material properties and subtle surface changes. This rich spectral information makes hyperspectral imaging highly useful for applications such as land monitoring, vegetation analysis, environmental assessment, and change detection.

1.2.2 Bi-Temporal Hyperspectral Image Change Detection

Bi-temporal hyperspectral image change detection refers to the process of identifying changes between hyperspectral images acquired at two different time instances over the same geographical area. The objective is to detect regions where significant spectral or spatial changes have occurred. This process is important in urban monitoring, land-use analysis, agriculture, and disaster assessment.

1.2.3 Change Vector Analysis (CVA)

Change Vector Analysis (CVA) is a classical spectral change detection method used to measure the magnitude of change between two temporal images. It computes the Euclidean distance between spectral vectors of corresponding pixels in the two images. A larger change magnitude indicates a higher probability of change. In this project, CVA is used as a preprocessing technique to enhance change-related information before deep learning-based classification.

1.2.4 Siamese Deep Learning Framework

A Siamese deep learning framework consists of two identical neural network branches that share the same architecture and weights. Each branch processes one of the bi-temporal hyperspectral images to extract spectral-spatial features. The learned features are then compared to identify differences between the two time instances. This architecture is effective for change detection because it can learn subtle spectral and spatial variations automatically.

1.2.5 Change Probability Map

A change probability map represents the likelihood of change for each pixel in the study area. It is generated by the deep learning model based on the extracted spectral-spatial features. Higher probability values indicate a greater possibility of change. This map provides a detailed visualization of potential changed regions.

1.2.6 Binary Change Map

A binary change map is the final output of the change detection framework, where each pixel is classified as either changed or unchanged. It is generated by applying a suitable threshold to the probability map. This binary representation helps in clearly identifying changed regions for further analysis.

1.2.6 Evaluation Metrics:

To assess the performance of building footprint extraction models, the following metrics are commonly used:

- **Precision:**
  Precision measures the proportion of correctly predicted building pixels among all predicted building pixels. A high precision indicates fewer false positives.
- **Recall:**
  Recall measures the proportion of actual building pixels that are correctly identified. A high recall indicates fewer missed detections.
- **Intersection over Union (IoU):**
  IoU refers to how much the predicted mask overlaps the ground truth mask. It is the commonly used metric for evaluating the accuracy of segmentation.
- **F1-Score:**
  F1-Score is the harmonic mean of precision and recall metrics, providing a balanced evaluation of model performance.
- **Accuracy:**
  Accuracy represents the overall correctness of predictions, both regarding buildings and non-buildings.
- **Area Under Curve (AUC):** Evaluates the model's ability to distinguish changed and unchanged pixels.
- **Dice Coefficient:** Measures overlap between predicted and reference changed regions.

1.3 PROBLEM STATEMENT:

The increasing availability of hyperspectral remote sensing data has led to new possibilities of monitoring the changes in the earth's surface throughout time. Bi-temporal hyperspectral images have ample spectral information useful for distinguishing changes in land cover and other environmental features. The problem, however, is the difficulty in change detection because of the highly dimensional and spatially complex nature of hyperspectral data. The manual approach in interpreting large amounts of data is cumbersome, inefficient, and not suitable in most applications.

The existing techniques for hyperspectral change detection include change vector analysis (CVA), Spectral Angle Mapper (SAM), and IR-MAD. These techniques rely on identifying the differences in spectra in bi-temporal images for detecting changes. Nevertheless, CVA, SAM, and IR-MAD suffer from various limitations such as illumination and atmosphere-induced artifacts. They often lead to erroneous results due to the problems in hyperspectral data.

Even though deep learning techniques have proved promising in analyzing hyperspectral images, the accurate detection of spectral-spatial differences is still difficult because of the nature of hyperspectral images. An efficient and effective system capable of learning valuable spectral-spatial features and providing accurate pixel-wise change maps is required. This research aims at addressing the challenges through a proposed deep learning framework using Siamese networks and CVA for pre-processing.

1.3.1 Objectives:

- To design an effective Siamese neural network approach for change detection in bi-temporal hyperspectral images. It will help to distinguish between changed and unchanged pixels in bi-temporal hyperspectral images.
- To use CVA-based preprocessing methods for maximizing spectral contrast. This will help in emphasizing the change information prior to extracting features and classifying pixels.
- To extract spectral-spatial features from bi-temporal hyperspectral images. These spectral-spatial features will enhance the capability of the model to detect surface changes in bi-temporal hyperspectral images.
- To create pixel-level change probability maps and binary change maps. It will allow accurate location and visualization of change information in the study area.

- To analyze the proposed approach in terms of quantitative and qualitative results. Various metrics such as accuracy, F1-score, Intersection Over Union (IoU), and Area Under Curve (AUC) are used.

1.3.2 Outcomes:

- An effective Siamese neural network-based deep learning system was designed for hyperspectral change detection. The designed model effectively extracted spectral-spatial features from bi-temporal hyperspectral datasets.
- A preprocessing step based on CVA effectively increased the difference in spectra at two different points in time. This ensured that effective input is obtained.
- A precise map of change probability and binary change detection were obtained. The maps precisely marked regions of change and no change in the hyperspectral dataset.
- The proposed model gave reliable results in terms of change detection, and there was very little to no noise in the results.
- The performance of the system was evaluated using various measures of evaluation. The comparison of spectra and correlation analyses confirmed the effectiveness of the system.

1.4 SOCIAL APPLICATIONS OF PROPOSED WORK:

- **Support for Environmental Monitoring:**
  The proposed change detection approach will assist in monitoring the changes that take place in forests, water bodies, vegetation, etc. This will assist environmental organizations in identifying the changes in the ecology and implementing conservation strategies.

- **Effective Change Detection on Large Scale:**
  The developed approach is efficient in processing large amounts of hyperspectral data through remote sensing. This makes it useful for the continuous monitoring of large geographical areas.
- **Higher Accuracy in Change Detection:**
  The use of the proposed deep learning approach enhances the accuracy of detecting changed and unchanged areas. This leads to accurate change maps.
- **Saves Human Efforts and Enhances Efficiency:**
  Through automation, the manual process involved in analyzing the multi-temporal images will be eliminated. This will reduce human efforts and save time for the analysis.
- **Framework for Future Studies:**
  The proposed framework provides a solid foundation for future studies in hyperspectral imagery. This may lead to intelligent geospatial applications.

1.4.1 APPLICATIONS:

- **Land Use and Land Cover Change Detection:**
  The proposed system will help in detecting any changes that occur in the land use and land cover categories over a period of time. It will be helpful for urban planning, agriculture monitoring, and sustainable resource management**.**
- **Disaster Management and Damage Estimation:**
  The proposed model will be helpful for detecting the change due to natural calamities such as floods, landslides, forest fires, etc. The changes detected will be helpful for assessing damages and formulating plans for mitigation.
- **Change in Agricultural Fields:**
  The proposed model will be able to detect changes in crop health, seasonality, and agricultural land. This information will assist farmers and organizations in improving their productivity and resource efficiency.
- **Change Detection for Urban Expansion and Infrastructure:**
  The proposed methodology will be able to detect any changes that take place in urban expansion and infrastructure development.

- **Environmental and Climate Change Studies:**
  The proposed framework will be helpful in conducting environmental and climatic changes studies in the long run.

1.5 SUMMARY:

In this study, we have designed a powerful Siamese-based deep learning approach to conduct change detection using bi-temporal hyperspectral imagery. For pre-processing of the input data, we make use of a preprocessing method named "Change Vector Analysis" (CVA). The proposed framework is capable of learning the spectral-spatial feature representation and obtaining precise change probability maps and binary change maps for detecting changed areas. For the assessment of the proposed system, we used the following quantitative measurements: accuracy, precision, recall, F1 score, IoU, dice similarity coefficient, and AUC score. We also applied several qualitative measures: Spectral Signature Matching and Correlation. Based on experimental findings, we proved that the performance obtained by our proposed system is both reliable and effective in bi-temporal hyperspectral image change detection. This system can contribute to real-world applications including but not limited to land-use management, agriculture, environment surveillance, disaster response and management, and urban growth analysis.

CHAPTER –2

REVIEW OF LITERATURE

The current chapter offers a literature review of previously conducted research papers dealing with the problem of automatic extraction of building footprints from aerial and satellite imagery. Different techniques relying on image processing, machine learning and deep learning are reviewed and the findings of the reviewed research papers are discussed with respect to the proposed methodology.

2.1 DESCRIPTION OF EXISTING SYSTEMS:

Land-Use/Land-Cover Change Detection Based on a Siamese Global Learning Framework for High Spatial Resolution Remote Sensing Imagery

Authors: Qiqi Zhu, Xi Guo, Weihuan Deng, Qingfeng Guan, Yanfei Zhong, Liangpei Zhang, Deren Li

Year: 2022

Observations:

The authors in this paper have suggested a semantic change detecting framework called Siamese Global Learning (Siam-GL) of bi-temporal remote sensing images. The framework consists of a Siamese encoder - decoder structure with shared weights, to obtain representative features of multi-temporal images. To deal with class imbalance, a global hierarchical sampling mechanism was applied and to minimize the impact of unchanged background areas, a binary change mask was applied. The suggested approach has a high level of performance on various datasets. The framework however incurs high computational complexity and needs a huge number of labelled training data which can limit its use in resource-constrained settings. [1]

A Hybrid Siamese Network with Spatiotemporal Enhancement and Two-Level Feature Fusion for Remote Sensing Image Change Detection

Authors: Liangliang Yan, Jie Jiang

Year: 2023

Observations:

HSSENet was proposed in this study as a hybrid architecture of a Siamese network, which uses both CNNs and Transformer-based models to detect changes. The system consists of a spatiotemporal enhancement unit to reduce pseudo changes and high-frequency noises, and also utilizes a two-tier feature fusion unit to refine the boundaries. It outperformed previous systems in terms of accuracy and efficiency. The drawback of this model is that it demands a lot of computational power during training time. [2]

Bitemporal Remote Sensing Image Change Detection Network Based on Siamese-Attention Feedback Architecture

Authors: Hongyang Yin, Chong Ma, Liguo Weng, Min Xia, Haifeng Lin

Year: 2023

Observations:

SAFNet was proposed by the authors in this research paper. This is a Siamese attention feedback network that performs bi-temporal remote sensing image change detection tasks. This network consists of three modules, namely global semantic, temporal interaction, and feature refinement. This framework attained remarkable performance metrics on benchmark datasets. Nevertheless, the framework's complexity increases its implementation challenges. [3]

Dual Learning-Based Siamese Framework for Change Detection Using Bi-Temporal VHR Optical Remote Sensing Images

Authors: Bo Fang, Li Pan, Rong Kou

Year: 2019

Observations:

This study suggested a novel change detection framework called Dual Learning-based Siamese Framework (DLSF). The model combines the advantages of domain translation and Siamese learning to decrease spectral differences between unchanged regions and enhance change recognition. The dual learning technique enables the suppression of domain shifts and enhances the accuracy of change detection. Nevertheless, this model has a relatively complicated structure and requires plenty of labeled data to train the model. [4]

A Multi-Dimensional Deep Siamese Network for Land Cover Change Detection in Bi-Temporal Hyperspectral Imagery

Authors: Seyd Teymoor Seydi, Reza Shah-Hosseini, Meisam Amani

Year: 2022

Observations:

In this paper, the authors have introduced a deep auto-Siamese network for multidimensional change detection in hyperspectral images. The proposed framework makes use of Otsu and DTW for training data set generation automatically, after which multi-dimensional CNN classification is performed. This framework made efficient use of spectral as well as spatial information to yield high accuracy along with reduced error rates. Nevertheless, the proposed framework has some limitations concerning hyperspectral noise, atmosphere effects, and computational issues with high dimensions. [5]

Traditional Hyperspectral Change Detection Methods (CVA / IR-MAD / SAM)

Authors: Various Researchers

Observations:

Conventional hyperspectral change detection approaches like change vector analysis (CVA), iteratively reweighted multivariate alteration detection (IR-MAD), and spectral

angle mapper (SAM) have found wide applications in hyperspectral image sequence change detection tasks. The simplicity and effectiveness of these techniques make them a reliable choice. Nevertheless, they are vulnerable to noise, illumination variations, and inaccurate registrations. In addition, they have low ability to learn complicated spatial-spectral relationships, thereby decreasing their performance in practice. This explains why deep learning change detection models are needed. [6]

2.2 SUMMARY OF LITERATURE STUDY

However, the reviewed literature indicates a gradual transition from conventional spectral based change detection techniques to more accurate and sophisticated deep learning based Siamese techniques. For example, conventional techniques such as Change Vector Analysis (CVA), Spectral Angle Mapper (SAM), and IR-MAD had been commonly used for spectral change detection in multi-temporal hyperspectral images. Although the methods had been quite easy to implement and compute, the accuracy levels remained poor as the algorithms had high sensitivities to noise, varying illumination, and complicated scenes.

In 2019, a new dual learning based Siamese framework was designed for bi-temporal change detection in very high-resolution remote sensing imagery. The algorithm incorporated the use of domain translation and Siamese learning to minimize spectral differences in unchanged areas and enhance change discrimination accuracy. The method had better accuracy, but the approach was characterized by a complicated computation process that necessitated labeled training data for efficiency [1]. On the other hand, another algorithm known as the Siam-GL had been developed in 2022 for semantic change detection in high-resolution remote sensing imagery. The technique mitigated sample imbalances through global hierarchical sampling and minimized misclassification via binary change masks. Nevertheless, the technique entailed high computation complexity [2].

Subsequent research in 2022 addressed the issue of change detection using a multivariate deep Siamese network in hyperspectral images. The use of an

automatically generated training sample and multidimensional convolution enabled the effective use of spectral spatial data in achieving high accuracy and minimal false positive results. However, issues related to hyperspectral noise and atmospheric conditions posed difficulties [3]. In 2023, sophisticated Siamese frameworks such as HSSENet and SAFNet were developed. HSSENet incorporated a transformer-enabled spatiotemporal enhancement technique and two-level feature fusion for enhanced border definition and pseudo-change suppression, while SAFNet utilized an attention feedback mechanism and feature enhancement module for improved structural integrity. These models demonstrated improved accuracy in change detection, though at the expense of increased computational intensity and complexity of the architecture [4], [5].

From the above discussion, it can be observed that change detection algorithms have come a long way from spectral-based approaches towards deep learning Siamese frameworks. While contemporary techniques have exhibited improved accuracy, some challenges still pose barriers, such as complexity in computations, noise vulnerability, extensive training data, and hyperspectral redundancy. These drawbacks make a case for developing an efficient and reliable framework for bi-temporal hyperspectral image change detection.

Table 2.1: Literature Survey Summary

Sno	Title and Publication Details	Algorithms or Technologies Used	Dataset	Purpose	Remarks
1	Dual Learning-Based Siamese Framework for Change Detection Using Bi-Temporal VHR Optical Remote Sensing Images (Bo Fang et al., 2019)	Dual Learning Framework, Siamese CNN	VHR Optical RS Images	Detect changes in bi-temporal images	Change detection performance enhanced; computationally intensive and requires labeled data.

2	Land-Use/Land-Cover Change Detection Based on a Siamese Global Learning Framework (Qiqi Zhu et al., 2022)	Siamese Encoder–Decoder, Global Hierarchical Sampling	High Spatial Resolution RS Images	Semantic change detection	Can manage class imbalances effectively; however, computationally intensive.
3	A Multi-Dimensional Deep Siamese Network for Land Cover Change Detection in Bi-Temporal Hyperspectral Imagery (Seydi et al., 2022)	Multi-Dimensional Siamese CNN, DTW, Otsu Thresholding	Hyperion Hyperspectral (Farmland-1, Farmland-2)	Hyperspectral change detection	Excellent OA and few false alarms; however, susceptible to noise and atmospheric interference.
4	RSCDNet: A Robust Deep Learning Architecture for Change Detection From Bi-Temporal High Resolution RS Images (Deepanshi et al., 2023)	Encoder–Decoder, Self-Attention, GL-ASPP	Staza-Tisadob, Onera, LEVIR, WHU	Pixel-wise change detection	F1-score is high; localization is effective; however, large amount of training data needed.
5	A Hybrid Siamese Network With Spatiotemporal Enhancement and Two-Level Feature Fusion (Yan & Jiang, 2023)	Hybrid Siamese CNN + Transformer, STEM, TL-FFM	LEVIR, WHU, DSIFN	Accurate boundary-aware change detection	Enhanced performance and efficiency; however, complex model architecture.
6	Bitemporal Remote Sensing Image Change Detection Network Based on Siamese-Attention Feedback Architecture (Yin et al., 2023)	Siamese Attention Feedback Network, GSM, TIM, FRM	Remote sensing benchmark datasets	Structural change detection	Edge detection and target completeness improved; however, increased training complexity.

2.3 SOFTWARE REQUIREMENT SPECIFICATION

2.3.1 Purpose

The goal of this project is to create an efficient deep learning architecture of the Siamese type to detect change in bi-temporal hyperspectral imagery. The goal of the system is to perform bi-temporal hyperspectral imagery analysis to identify change and non-change areas.

2.3.2 Scope

The system is designed to process bi-temporal hyperspectral imagery for detecting changes. The process involves preprocessing, extracting features in spectral-spatial domains, enhancing the results using CVA, deep learning-based change classification, generating change probability maps, and evaluating the performance of the proposed system. The application areas of the system include environmental monitoring, agriculture, urban planning, and disaster management.

2.3.3 Intended Audience

The target audience of this system consists of remote sensing specialists, geospatial analysts, environmental organizations, urban planners, agricultural professionals, and disaster management organizations. The project can also be used as a source of information by students and scientists researching hyperspectral imagery and deep learning architectures.

2.3.4 Functional Requirements

- FR1: The system will load and preprocess bi-temporal hyperspectral datasets.
- FR2: The system will conduct spectral normalization and CVA preprocessing.
- FR3: The system will extract spectral-spatial features via Siamese deep learning network.
- FR4: The system will classify pixels to either changed or unchanged classes.

- FR5: The system will produce change probability and binary change maps.
- FR6: The system will present RGB compositions, spectral profiles, and change detection output.
- FR6: The system will assess accuracy through metrics like accuracy, precision, recall, F1 score, IoU, Dice, and AUC.

2.3.5 Non-Functional Requirements

- NFR1: The system needs to ensure accurate and robust change detection outcomes.
- NFR2: The system needs to efficiently handle hyperspectral datasets with high dimensions.
- NFR3: The system needs to have scalability capability for large remote sensing images.
- NFR4: The system needs to be robust against noise and illumination differences.
- NFR5: The system needs to offer visually interpretable results.
- NFR6: The system needs to be developed with ease of use and maintenance using Python and deep learning libraries.

2.4 SUMMARY

This chapter provided an extensive analysis of available approaches to building footprints detection, including classical machine learning algorithms as well as contemporary deep learning models. It outlined both advantages and disadvantages of each approach, thus emphasizing the problem complexity and the need for an optimal model. In addition, it set out the requirements to the proposed method, thus laying the basis for its development and comparison with other approaches.

CHAPTER-3

METHODOLOGY

This chapter focuses on the methodology that has been employed for performing the bi-temporal hyperspectral images' change detection. The data sets involved in the process, as well as pre-processing stages carried out and the complete framework established to enable accurate identification of changed/unchanged areas, will be discussed.

3.1 DESIGN METHODOLOGY AND ARCHITECTURE

In this part, the architecture of the proposed Siamese deep learning system for bi-temporal hyperspectral images' change detection is introduced. The purpose of developing this system is to efficiently detect spectral and spatial changes that occurred within the same object between two different acquisitions of hyperspectral images. In the first step, bi-temporal hyperspectral data sets are employed as the input data of the proposed framework. They are provided in ENVI file formats. Before using the dataset, they are preprocessed via ROIs extraction, normalization, noise removal, and robust scaling.

The next step is feeding the hyperspectral images of the same object in two temporal frames to two Siamese neural networks of deep learning with shared weights and feature consistency. In order to extract features, the two Siamese branches simultaneously extract spectral and spatial features of both images. To increase change information, CVA and spectral differencing algorithms are applied. After extracting the difference features, they are passed through a classification network to identify the pixels which are classified into changed and unchanged categories.

The proposed framework provides change probability maps as well as binary change maps that make the changes detected easily visible. Various measures have been applied in order to test how effectively the framework can detect changes including Overall Accuracy, Precision, Recall, F1-score, Dice coefficient, IoU, and AUC. Apart from the above measures, various qualitative analyses have been conducted in order to

prove that the proposed change detection framework is reliable and robust. These include the use of RGB composite images, spectral signatures, ROC curves, and correlations graphs.

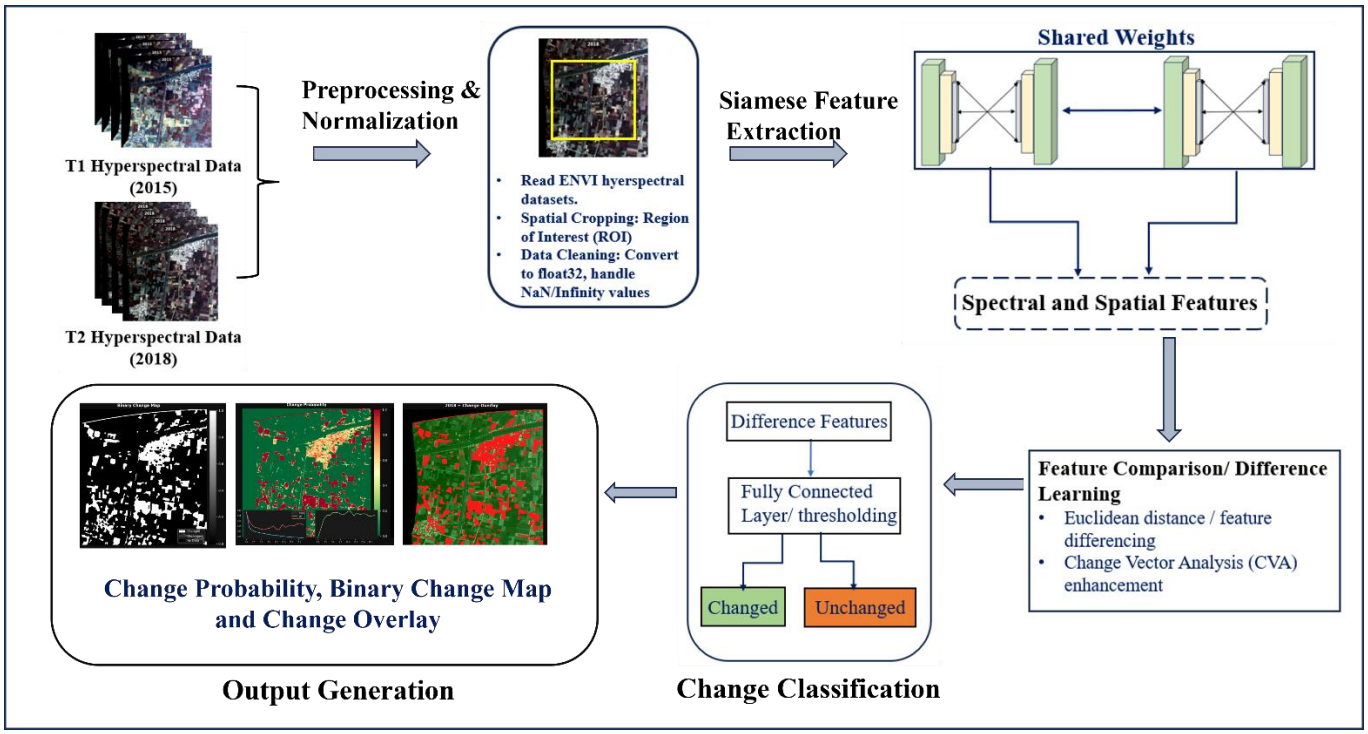


***Fig 3.1:** Architecture diagram*

3.1.1 Dataset Description

The dataset that will be utilized in this project involves two sets of hyperspectral images that have been captured from the AVIRIS-NG sensor for the years 2015 and 2018. The chosen study area involves the geographical area of Madduru and the nearby surroundings. These include agricultural areas, water resources, and other diverse forms of land uses.

The datasets used are hyperspectral images in ENVI (.hdr) format. Here, the images are considered as three-dimensional cubes comprising both spatial and spectral information in the form of pixels and spectral bands. Every dataset comprises spatial dimension 5614 × 707 pixels and spectral dimension having 372 spectral bands, respectively.

T1 indicates the image captured during the year 2015 which shows the initial scenario of the land surface, whereas T2 indicates the image captured during the year 2018 showing the changes happened during the period because of various factors like vegetation growth, land use changes, etc. As hyperspectral imagery has the ability to record spectral information very accurately, therefore, it becomes possible to detect minute changes in the materials.

In order to enhance computational speed and consider only the important part, an ROI is selected from the entire dataset. The ROI chosen is indicated in the above image by a box that marks the location from the dataset where both agricultural fields and other elements exist. This part will be used for pre-processing, feature extraction, and change detection.

In general, the AVIRIS-NG bi-temporal hyperspectral dataset can be considered a very effective and useful source of spectral-spatial data to detect changes with time.

Steps to Download Data Set:

Step-1 :Open AVIRIS using the following link:

https://aviris.jpl.nasa.gov/data/get_aviris_data.html

After that, click on the link "Get AVIRIS Data". This will open the AVIRIS Data Products Portal in a new tab.

Step-2: From the AVIRIS Data Product portal, choose the required dataset from the Layers window on the left-hand side of the interface. Select the AVIRIS-NG layer to view the coverage for the required area on the map.

Step-3: Once you have selected the AVIRIS-NG layer, click on AVIRIS-NG Coverage to see the coverage map. After that, click on the location point icon indicated next to the layer to get zoomed into the selected coverage area and choose the available hyperspectral images.

Step-4: Use the timeline filter present at the bottom of the web page to choose the required acquisition time. Change the start and end date range, to see the AVIRIS scenes taken during your desired time interval.

Step-5: Click the AVIRIS scene available on the map indicated in blue color. After clicking the selected AVIRIS scene, the link will be displayed at the top of the map and showing three different download options. These options include the download links for Radiance, Quicklook, and Reflectance datasets, as indicated in the figure.

3.1.2 Data Preprocessing and Normalization

Data preprocessing plays a crucial role in the developed bi-temporal hyperspectral image change detection framework because raw data usually includes some noise, invalid values, and other superfluous data. Preprocessing helps significantly increase the quality of data and provides reliable features used in change detection processes.

At the initial stage of change detection, two hyperspectral datasets from the years 2015 and 2018 are loaded from the file format .hdr of software ENVI with the help of the required libraries. The point is that hyperspectral data are extremely big in size and consist of several hundred spectral bands; therefore, there is a need to choose a Region of Interest (ROI), which will include only the study area.

Next, the ROI is transformed into the float32 format for further processing. Any invalid values such as NaN or infinity are detected and substituted because they may cause an error in further work. Robust techniques of normalization such as percentile-based clipping are applied in order to achieve the most consistent results and eliminate any influence of the different illuminations.

3.2 DESCRIPTION OF ALGORITHMS

The proposed bi-temporal hyperspectral image change detection framework is designed to identify temporal variations between hyperspectral images captured at two different time instances. The complete methodology combines preprocessing, deep

feature extraction, spectral change analysis, classification, and evaluation steps to achieve accurate change detection. Each algorithm used in the framework plays a specific role in improving the quality and reliability of the final results.

3.2.1. Data Preprocessing Algorithm:

The hyperspectral images typically possess high spectral dimensionality data in addition to noise, missing data, and redundant backgrounds. It is important to perform preprocessing to ensure that input data quality is enhanced before the use of change detection methods.

Firstly, the bi-temporal hyperspectral images are imported from ENVI files. As the entire scene holds a huge amount of data, it is important to extract the Region of Interest (ROI). The computational cost is significantly reduced, and the framework focuses on the most critical part of the research area. Subsequently, the image data is transformed into numeric data. Any invalid numbers like NaN or infinity are detected and rectified to prevent instabilities in calculations.

Normalization enhances spectral consistency and minimizes any impact due to illumination differences. Normalization transforms pixel values into a common scale so that the model can learn more efficiently.

Normalization Formula

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad \textbf{(1)}$$

Where:

- X= original pixel value
- X_{min}, X_{max}= minimum and maximum values

Purpose:

- Removes noise and invalid values
- Improves spectral consistency
- Reduces processing complexity

3.2.2. Siamese Deep Learning Algorithm

The major framework used for the model development of this project is that of a Siamese deep learning network, which is very appropriate when dealing with two images that are temporal in nature. In this case, the Siamese model has two branches that are identical, with similar architectures and shared weights.

This implies that all the input images are processed equally through each of the branches. In this model, the first branch works on the image obtained from the year 2015, while the second works on the hyperspectral image obtained in 2018. Both the branches extract the spectral and spatial features.

It is important to note that spectral features refer to the materials reflectance in terms of different spectral bands. On the other hand, spatial features relate to structures and textures. Given the number of spectral bands in the images, this model is very appropriate when extracting features automatically.

Siamese Feature Extraction Formula

For two temporal hyperspectral images $I_1$and I_2:

$$F_1 = f(I_1; \theta)$$
$$F_2 = f(I_2; \theta) \quad \textbf{(2)}$$

Where:

- I_1, I_2= input temporal images
- $f(.)$= feature extraction function
- θ= shared network parameters
- F_1, F_2= extracted deep feature vectors

Purpose:

- Learns spectral-spatial information
- Provides consistent feature extraction

- Handles high-dimensional hyperspectral data

3.2.3 Change Vector Analysis (CVA) Algorithm

The feature extraction process leads to using Change Vector Analysis (CVA) to calculate the change vector magnitude between two temporal hyperspectral images. The CVA technique is a good spectral analysis tool which calculates the spectral differences between vectors at corresponding pixel positions.

This implies that the spectral difference for all pixels in the bands will be calculated. High values mean high probability of change, while low values suggest the absence of change. The application of CVA helps detect land surface changes which cannot be detected using RGB images.

CVA Formula

$$\boldsymbol{CVA = \sqrt{\sum_{i=1}^{n}(X_{2i} - X_{1i})^2}} \qquad \textbf{(3)}$$

Where:

- X_{1i}= spectral value of band $i$in 2015
- X_{2i}= spectral value of band $i$in 2018
- n= number of spectral bands

Purpose:

- Measures spectral change magnitude
- Highlights change boundaries
- Enhances change information

3.2.4. Feature Differencing and Euclidean Distance

Other than CVA, other methods of comparing the features generated through Siamese network include feature differencing and Euclidean distance. This is useful in distinguishing the spectral and spatial differences that may exist between the two images in time.

Euclidean distance finds out the difference between the vectors. This is further used for evaluating the presence of temporal change.

Euclidean Distance Formula

$$D = \sqrt{\sum_{i=1}^{n} (F_{2i} - F_{1i})^2} \quad \textbf{(4)}$$

Where:

- F_1, F_2= deep feature vectors

Purpose:

- Measures deep feature variation
- Improves discrimination between changed and unchanged pixels

3.2.5. Change Classification Algorithm

The differential features that have been extracted by CVA and feature comparison using Siamese are fed into the classifier module. Here the pixels are classified based on whether they belong to the changed region or unchanged region.

This can be achieved either using a thresholding technique or using fully connected classifier layers. If the change value is greater than the threshold value, then the pixel will be considered a changed pixel.

Threshold Classification Formula

$$C(x, y) = \begin{cases} 1, & \text{if } D(x, y) > T \\ 0, & \text{if } D(x, y) \leq T \end{cases} \quad \textbf{(5)}$$

Where:

- $D(x, y)$= change value at pixel (x, y)
- T= threshold value

Purpose:

- Performs final decision making

- Generates binary change map

3.3 METHODOLOGY OF PREVIOUS STUDIES

The methodology adopted for this research comprises thorough experiments that would assess the effectiveness of the proposed Multimodal Fusion Transformer (MFT) in remote sensing image classification. In particular, the study is concerned with multimodal land-cover classification through fusion of hyperspectral imagery and other complimentary data sources like LiDAR, SAR, MSI, and DSM. The experiment entails feature extraction, multimodal token fusion, transformer based cross-patch attention, and classification of various types of land covers.

3.3.1 Study 1 - Multimodal Remote Sensing Image Classification Using Multimodal Fusion Transformer (MFT)

This methodology provides a complete approach towards multimodal remote sensing image classification through the Multimodal Fusion Transformer (MFT). The approach has been made such that it can effectively combine the HSI images with other modalities like LiDAR, SAR, MSI, and DSM for classification of the land cover. The proposed approach makes use of convolutional feature extraction, tokenization, and cross-patch transformer based attention mechanism.

Data collection:

The multimodal remote sensing datasets are obtained from the following publicly accessible benchmark databases:

University of Houston Database (HSI + LiDAR + MSI)

Trento Database (HSI + LiDAR)

MUUFL Gulfport Database (HSI + LiDAR)

Augsburg Database (HSI + SAR + DSM)

The datasets collected consist of:

Images acquired using hyperspectral imaging technology having multiple spectral bandsOther modality information such as elevation in case of LiDAR, backscatter for SAR and height for DSM Land cover annotations provided by ground truthing

Some of the various types of land cover classes include:

- Buildings
- Roads
- Vegetation
- Water Bodies
- Bare Soil
- Residential Lands

Preprocessing

The data is first preprocessed before the feature extraction and classification process.

Data preprocessing involves:

- Denoising of spectral bands
- Normalization of pixels
- Alignment of multimodal images
- Geometry correction
- Extraction of image patches

For each pixel location (i, j):

- An HSI spectral-spatial cube of size $k \times k \times B$is extracted
- A corresponding multimodal patch of size $k \times k \times C$is extracted

Where:

- k= patch size (typically 11 × 11)
- B= number of HSI bands
- C= number of modality channels

This ensures spatial correspondence between HSI and complementary data patches.

HSI Feature Extraction Using CNN

To extract spectral-spatial discriminative features from HSI patches, a lightweight convolutional neural network is used.

The HSI patch is first processed through:

- 3D Convolution Layer
- Batch Normalization (BN)
- ReLU Activation

This captures spectral and spatial information jointly.

Then:

- HetConv2D layer is applied for efficient spatial feature extraction

The feature extraction process is:

$$X_{in} = Conv3D(X_H) \quad \textbf{(6)}$$

$$X_{out} = Conv2D_{group}(X_{in}) \oplus Conv2D_{point}(X_{in}) \quad \textbf{(7)}$$

Where:

- X_H= HSI input patch
- X_{out}= extracted HSI feature map

This step reduces spectral redundancy while preserving important local information.

HSI Patch Tokenization

The extracted HSI feature maps are converted into transformer patch tokens.

Steps include:

- Flattening HSI feature maps
- Selecting important patch tokens using learnable tokenization weights

The process is:

$$X_H^{flat} = T(Flatten(X_{out})) \quad \textbf{(8)}$$

$$X_H^{patch} = X_H^{Wa} \cdot X_H^{Wb} \quad \textbf{(9)}$$

Where:

- X_H^{patch}= selected HSI patch tokens
- Wa, Wb= learnable weights

This reduces feature dimensionality and preserves the most informative spectral-spatial features.

Multimodal Feature Tokenization (LiDAR / SAR / DSM)

Complementary modality patches are processed separately to generate an external class token (CLS token).

Steps include:

- Conv2D operation
- Batch Normalization
- GELU activation
- Tokenization

The process is:

$$X_L^{conv} = T(Flatten(Conv2D(X_L^{patch}))) \quad \textbf{(10)}$$

$$X_L^{cls} = X_L^{Wa} \cdot X_L^{Wb} \quad \textbf{(11)}$$

Where:

- X_L^{patch}= multimodal input patch
- X_L^{cls}= generated external CLS token

This token captures elevation, structural, or complementary scene information.

Fusion of HSI and Multimodal Tokens:

The multimodal CLS token is concatenated with HSI patch tokens:

$$X^{HL} = [X_L^{cls} \parallel X_H^{patch}] \quad \textbf{(12)}$$

Then:

- Positional embeddings are added
- Dropout is applied to reduce overfitting

This enables the transformer to learn both global and local context jointly.

Multihead Cross Patch Attention (mCrossPA):

The core contribution of the proposed methodology is the Multihead Cross Patch Attention mechanism.

This module enables:

- Interaction between HSI spectral-spatial tokens
- Complementary multimodal token fusion
- Long-range contextual learning

Cross-attention process:

$$Q = X_L^{cls} W_q, K = X^{HL} W_k, V = X^{HL} W_v \quad \textbf{(13)}$$

Attention weights:

$$Z = softmax\left(\frac{QK^T}{\sqrt{h_d}}\right) \quad \textbf{(14)}$$

Cross-patch attention output:

$$CrossPA(X^{HL}) = Dropout(W_l ZV) \quad \textbf{(15)}$$

Where:

- Q= Query
- K= Key
- V= Value
- h_d= head dimension

This mechanism helps the model effectively fuse multimodal complementary information.

Study 1 Algorithm

Algorithm: Multimodal Fusion Transformer (MFT)-Based Land Cover Classification

Inputs: Hyperspectral Image (HSI), complementary modality data (LiDAR / SAR / MSI / DSM), ground truth labels

Outputs: Classified land-cover map with accurate class labels for each pixel

Methods:

1.Loading Multimodal Remote Sensing Data

- Load hyperspectral image (HSI) dataset.
- Load dataset of complementary modality data (LiDAR/SAR/MSI/DSM).
- Load labels for training/validation.

2. Data Pre-processing

- Remove noisy bands from HSI data.
- Normalize pixel intensity values.
- Perform geometric correction and spatial alignment of multimodal data.
- Georeference all modalities.
- Crop patches around each pixel coordinate.

3. Patch Extraction

a. Extract spectral-spatial patches of size 11x11xB from HSI data.

b. Extract complementary multimodal patches of size 11x11xC.

c. Ensure pixel-wise alignment between modalities.

4. Feature extraction of HSI using CNN

a. Input HSI patches into Conv3D layer to learn spectral-spatial features.

b. Apply Batch Normalization (BN).

c. Apply ReLU activation function.

d. Feed output to HetConv2D to effectively learn spatial features.

e. Generate compact HSI feature maps.

5. Tokenizing HSI Patches

a. Flatten generated HSI feature maps.

b. Tokenize HSI feature maps by flattening them further.

c. Select important tokens using learnable token weights.

d. Retain critical information while reducing feature redundancy.

6. Complementary Modality Tokenization (LiDAR/SAR/DSM/MSI)

a. Input multimodal patch into Conv2D layer.

b. Apply Batch Normalization (BN).

c. Apply GELU activation function.

d. Learn and generate external CLS token.

e. Gather elevation/structural/contextual information from complementary modalities.

7. Multimodal Token Fusion

a. Concatenate external CLS token with HSI patch tokens.

b. Add positional embeddings to all tokens.

c. Dropout to avoid overfitting.

d. Prepare fused tokens for encoder processing.

8. Multihead Cross Patch Attention (mCrossPA)

a. Calculate Query (Q) using external CLS token.

b. Calculate Key (K) and Value (V) based on multimodal tokens fusion.

c. Perform cross attention calculation.

d. Interaction learning among HSI and multimodal features complementarities.

e. Global context enhancement.

9. Transformer Encoder Steps

a. Layer normalization process.

b. Process tokens with multihead cross patch attention.

c. Residual connections.

d. Process via MLP block.

e. Semantic long-range interactions learning.

10. Classification of Land Cover

Pass fused CLS token to classifier module.

Apply softmax function.

Assign class label per pixel.

Creating Output

Produce classified land cover map.

Classifying pixels to generate output.

Recognizing different land covers like:

- Buildings
- Roads
- Vegetation
- Water bodies
- Bare soil

3.3.2 Study 2 – Multimodal Land-Cover Classification Using TGFNet

The proposed methodology offers a classification pipeline that automatically classifies land-cover types by combining Hyperspectral Image (HSI) data and LiDAR data

through the use of the designed network referred to as TGFNet. The methodology utilizes techniques such as spectral-spatial features learning, transformers, and multimodal data fusion for enhanced classification in remote sensing applications.

Data Acquisition

High-resolution multimodal remote sensing datasets are collected from benchmark datasets such as:

- Houston Dataset
- Trento Dataset

These datasets contain:

- Hyperspectral imagery with multiple spectral bands
- LiDAR elevation and spatial information
- Ground truth class labels

Images are categorized into:

- Training set
- Validation set
- Test set.

Data Pre-processing

Data preprocessing is employed to guarantee consistent data quality and improved model accuracy through the following steps:

- Removing noise in spectral bands
- Normalizing pixel intensity
- Alignment of spatial information between HSI and LiDAR datasets
- Image geometry correction
- Creating image patches (11 × 11)
- Image augmentation (resizing, rotating, flipping, cropping)

These steps guarantee:

- Pixel-level consistency among modalities
- Higher quality data
- Classification model robustness

Feature Reconstruction Using FRM

Feature Reconstruction Module (FRM) is applied for further improving the multimodal input features and removing the redundant information.

Major activities involved are:

- Decomposing HSI and LiDAR features through NMF
- Removing redundant and noisy features
- Reconstructing spectral and spatial features

Outputs include:

- Reconstructed HSI features (F_{HSI})
- Reconstructed LiDAR features (F_{LIDAR})
- Cross-channel fused features (F_{CCF})

Spectral Feature Extraction Using Transformer (TSFEM)

The Transformer-based Spectral Feature Extraction Module (TSFEM) extracts rich spectral and contextual information from hyperspectral inputs.

Major activities involved are:

- Dividing HSI patches into spectral tokens
- Flattening and embedding tokens
- Adding positional encodings
- Passing tokens through transformer encoder blocks
- Learning long-range spectral dependencies

Outputs include:

- Spectral attention feature maps

- Improved material discrimination
- Enhanced contextual understanding

This step improves spectral feature representation.

Spatial Feature Extraction Using Gist CNN (GSFEM)

The Gist-based Spatial Feature Extraction Module (GSFEM) extracts local spatial and structural information from LiDAR and fused inputs.

Major activities involved are:

- Applying 2D convolution layers
- Extracting spatial texture patterns
- Capturing elevation and shape information
- Using batch normalization and ReLU activation
- Refining features through residual blocks

Outputs include:

- Spatial feature maps
- Elevation-aware structural features
- Local texture representation

This improves land-cover boundary understanding.

Multimodal Feature Fusion

Features extracted from HSI, LiDAR, and reconstructed channels are fused to create a rich unified representation.

Major activities involved are:

- Combining HSI spectral features
- Combining LiDAR spatial features
- Combining cross-channel reconstructed features
- Learning inter-modal dependencies

Outputs include:

- Fused multimodal feature vector (F_{Fusion})
- Improved class separability
- Enhanced discrimination between similar classes

This step enables effective multimodal interaction.

Land-Cover Classification

The fused features are passed through the classification head to predict land-cover classes.

Major activities involved are:

- Passing fused features through fully connected layers
- Applying Softmax activation
- Predicting class probabilities
- Assigning final land-cover labels

Outputs include:

- Pixel-wise classified land-cover map
- Class probabilities
- Identification of classes such as buildings, vegetation, roads, and water

This step provides final classification results.

Output Generation:

Final classification outputs are prepared for visualization and GIS applications.

Outputs include:

- Classified land-cover maps for Houston and Trento datasets
- Per-class accuracy charts
- Confusion matrices
- GIS-compatible raster outputs

These outputs support:

- Urban planning
- Agricultural monitoring
- Environmental analysis
- Disaster management applications

3.4 SUMMARY

The methodology proposed in this paper utilizes the Transformer and Gist CNN Fusion Network (TGFNet) model for land-cover classification via automatic multimodal fusion of remote sensing data. First, hyperspectral and LiDAR remote sensing data sets are acquired and pre-processed through normalization, denoising, spatial registration, and patch extraction processes. Afterward, the Feature Reconstruction Module (FRM) helps in improving the input data by eliminating redundancy in the input features while keeping relevant information intact. The feature vectors are extracted from the refined data set by using the TSFEM that extracts long-range spectral dependency information. Simultaneously, the spatial features are extracted with the help of GSFEM that extracts spatial and structure-related patterns from the input features. The final extracted features are fused and fed into the classifier module to predict the land-cover classes using softmax function. Performance assessment and post-processing of the results are done by calculating Overall Accuracy, Average Accuracy, Kappa coefficient, Precision, Recall, and F1 score.

CHAPTER-4

RESULTS AND OBSERVATIONS

In this chapter, a detailed analysis is made on the output results generated by the developed transformer-based fusion framework for the hyperspectral and LiDAR data classification task. In particular, an examination is made regarding the performance level of the proposed architecture on various datasets including Houston and Trento, considering the efficiency of classification as well as the training process and class-wise performance. The framework makes use of advanced deep learning methods such as 3D-CNN, 2D-CNN, and transformers' encoders in order to utilize spectral-spatial information and boost cross-modal interaction capabilities.

4.1 STEP WISE DESCRIPTION OF RESULTS AND ANALYSIS

The results and their analysis are systematically done in a step-wise fashion for assessing the performance of the proposed model for fusion of hyperspectral and LiDAR data. The first step involves training the model with the prepared datasets and analyzing training and validation loss curves to ensure good convergence and stability. In the next step, the outputs from the model are produced in the form of classified maps that illustrate how the model has discriminated the different land cover types. Then comes the analysis part where performance metrics like overall accuracy, average accuracy, precision, recall, and F1-scores are calculated for understanding the performance of classification. In addition to this, the per class accuracy is also checked for assessing classification errors.

4.2 TGF-Net Model Output Classified Maps

The Transformer and Gist CNN Fusion Network (TGF-Net) is an innovative deep learning network that leverages the benefits of convolutional neural networks and transformers to efficiently incorporate hyperspectral and LiDAR data in multimodal remote sensing image classification. It makes use of the power of the 3D-CNN to

extract the spectral–spatial feature map from the hyperspectral image data and the 2D-CNN to learn the structural and spatial properties of LiDAR data. To integrate the two sets of features into one, it employs the transformer encoder to fuse the complementary information, allowing cross-modal interactions among the features via attention mechanisms and capturing long-range dependencies. Moreover, it employs feature reconstruction and gist-based spatial extraction techniques to further optimize and accentuate the informative elements of the features and minimize redundancies.

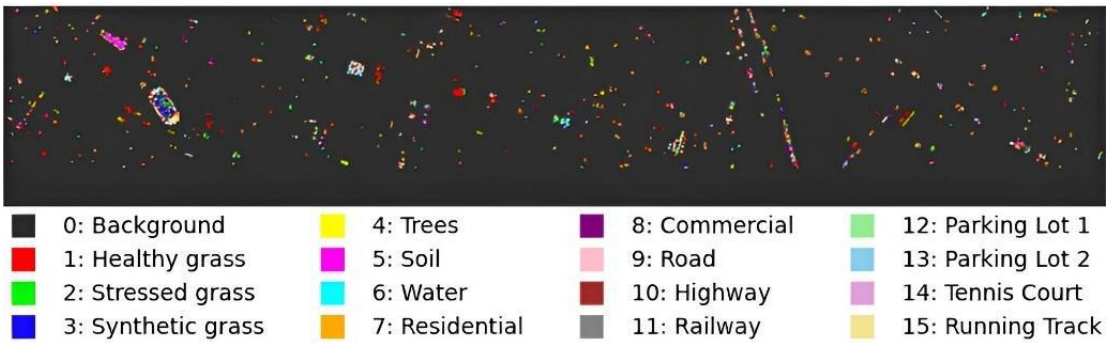


***Fig 4.1:** TGF-Net Results for Houston Dataset*

The output classified map generated by the TGF-Net model provides a visual representation of the predicted land-cover categories for each pixel in the input datasets. Different colors are used to denote various classes such as vegetation, buildings, roads, water bodies, and other surface types, allowing for easy interpretation of the classification results. The map clearly demonstrates the model's ability to capture both fine spatial details and complex spectral variations, resulting in well-defined boundaries between different classes. Regions with similar spectral characteristics are effectively distinguished due to the strong feature fusion capability of the model. Additionally, the presence of minimal noise and limited misclassification indicates the robustness and precision of the approach. Overall, the classified maps validate that the TGF-Net model successfully learns discriminative features and produces accurate, high-quality land-cover representations suitable for real-world remote sensing applications.

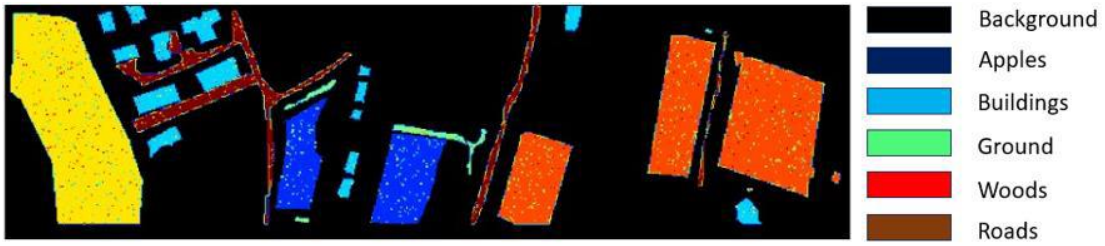


***Fig 4.2:** TGF-Net Results for Trento Dataset*

Classification map showing outputs for Trento Data Set shows the performance of the proposed TGF-Net model in accurately distinguishing various land cover types from an agricultural setting. In the output, each type of class is shown in a particular color, making it easier for the observer to understand the classification process. It is evident that the model performs extremely well in recognizing subtle spectral and spatial differences among the features in the image. This makes it possible for the TGF-Net to distinguish between similar types of vegetation such as apple orchards and vineyards.

Moreover, it can be seen that the classified image clearly demarcates regions from each other, thus showing that there has been marked improvement in discrimination of features through the fusion of hyperspectral and LiDAR data. The model is able to minimize any confusion regarding classes that have similar features, such as roads and bare ground, by using information about the elevation of the structure, thanks to the data generated through LiDAR. It can only be noted that some errors occur when there are changes in transitions, but this too occurs because there is similarity in spectral information.

4.3 TRENTO DATASET TRAINING AND LOSS

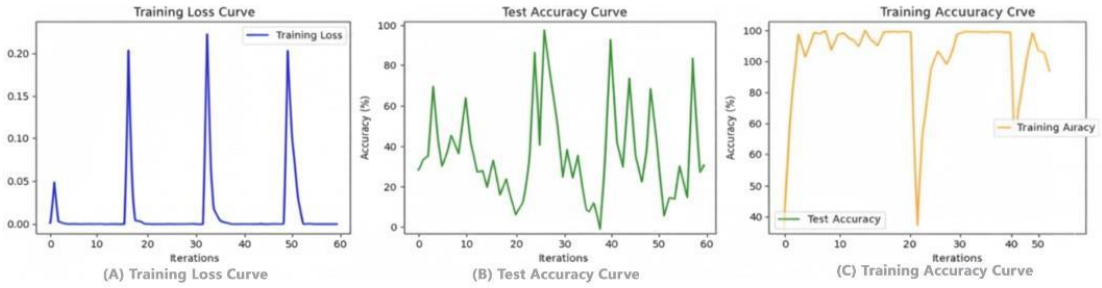


Fig 4.3: Trento dataset training and loss graphs

The following graph represents the different graphs of training and testing performance based on accuracy and loss changes of the model with regard to several iterations during the training process on the Trento dataset. Firstly, regarding the training loss graph, one could notice that the loss value was about 0.20 at first but then dramatically dropped down after several iterations until stabilization at 0.005 approximately. In other words, as a result of fast convergence, both the reconstruction error and classification errors were suddenly reduced. As for the test accuracy graph, one could see that the value of accuracy increased from 40% in the early stages up to 98-99%. There is a small variation in the first few iterations because of the nature of the data set and frequent validation updates. Likewise, the graph of training accuracy gradually rose from about 45 percent initially to nearly 100 percent towards the end, indicating very efficient learning with minimal chances of underfitting. In summary, it is evident that the model achieved stability, high classification accuracy, and excellent generalization on the Trento data set.

4.4 HOUSTON DATASET TRAINING AND LOSS

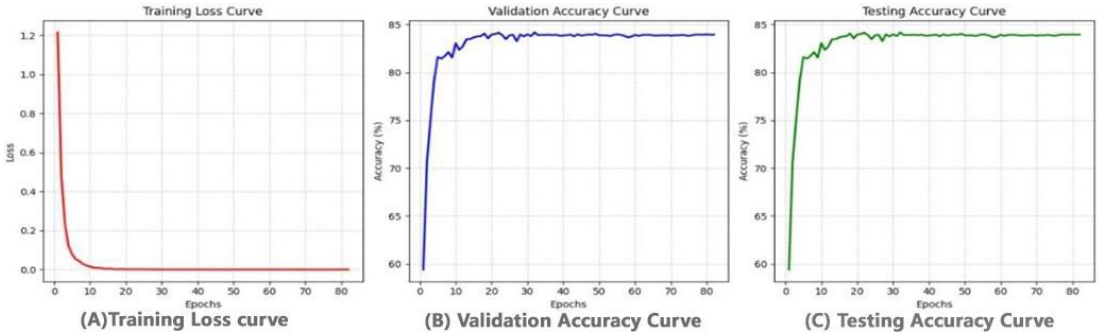


Fig 4.4: Houston dataset training and loss graphs

The plot below represents the learning and generalization curves for the proposed architecture on the Houston data set. From the training loss curve, it is observed that there is a steep decline in loss from about 1.2 during the early epochs to 0.01 at the 20th epoch after which it levels off, implying that the proposed model was able to minimize classification loss efficiently. Validation accuracy curve is a fast one during the initial 15-20 epochs, rising from about 60% to around 85%, followed by a stable trajectory in subsequent stages, demonstrating convergence and absence of overfitting very convincingly. Likewise, testing accuracy curve begins with approximately 60% accuracy, ending at up to 84-85% after the latter epochs, indicating consistency and generalization on unknown data. Efficient optimization, balanced training-validation learning, and reliable classification are some other aspects that can be derived from the aforementioned smoothly sloping curves from the Houston data set.

4.5 PER CLASS ACCURACY OF HOUSTON DATASET

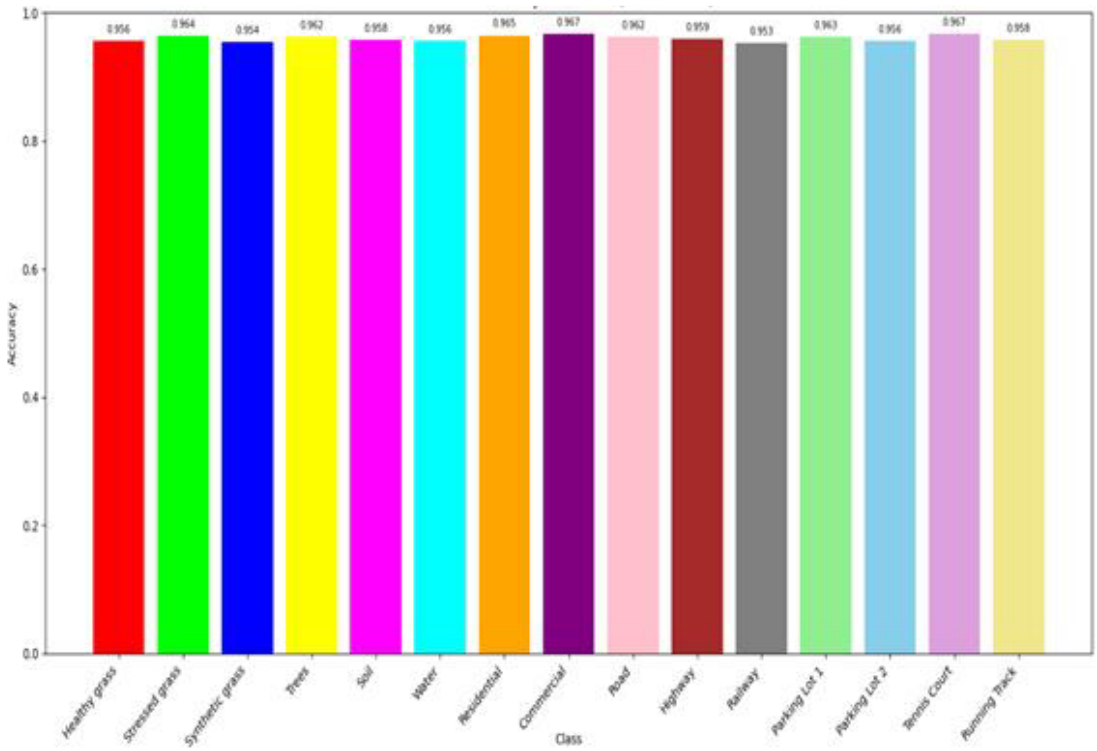


Fig 4.5: Per class Accuracy of Houston Dataset.

The above graph represents the accuracies distribution of each class among all 15 land cover classes in the Houston data set, and hence, the performance of the model can be evaluated according to different ground cover classes. Here, it is observed that the proposed model achieved very high accuracies for most classes with the range of values lying between 0.953 and 0.967. Among these, commercial area and tennis court classes got the first rank with an accuracy of 0.967. Even those classes that might be more difficult to recognize, such as Railway (0.953) and Synthetic grass (0.954), had excellent results, demonstrating how the subtle differences in spectra-spatial characteristics were recognized by the model. As can be observed, the variability in terms of the proposal's performance between the classes is minimal, which implies that the proposed transformer-based fusion architecture may provide a good balance between natural and artificial objects. Overall, all fifteen classes had very similar

results, thus ensuring the consistency of the proposed model for the difficult task of urban land cover classification from multisource information.

4.6 PER CLASS ACCURACY OF TRENTO DATASET

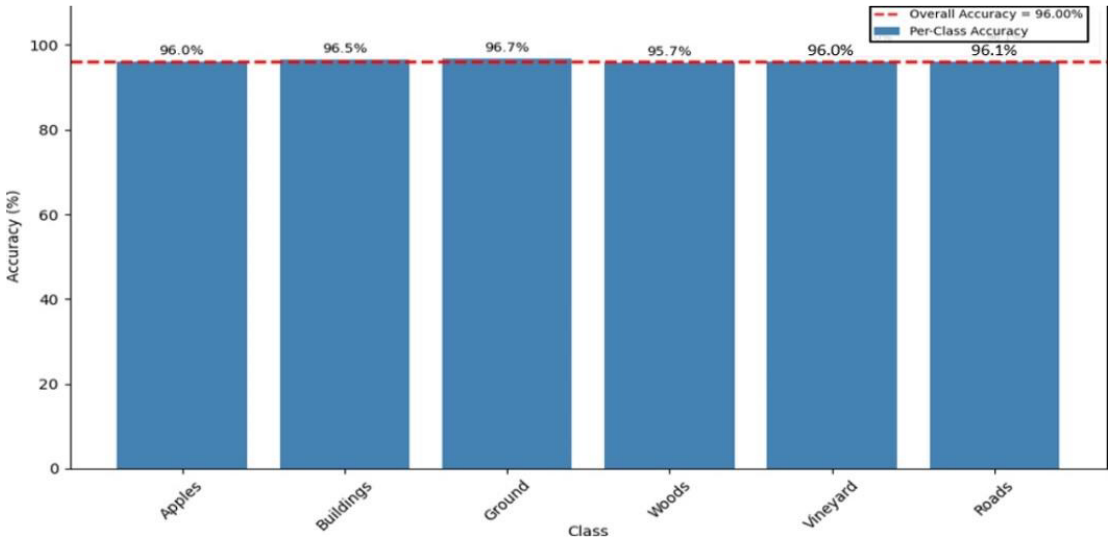


Fig 4.6: Per class Accuracy of Trento Dataset.

Table below is the analysis of the classification accuracy of Trento data set, which allows assessment of performance accuracy for all land cover types. The model's overall accuracy stands at 96.0%, indicating remarkable consistency and balance throughout the learning process of all the classes. Among the per-class accuracy, the following classes have a higher accuracy rate: Ground (96.7%) and Buildings (96.5%), together with several categories, including Woods (95.7%), Apples (96.0%), and Vineyard (96.0%), 35 all performing quite impressively. Even Roads, a class that has spectral resemblance to many other surfaces, shows a relatively high accuracy rate of 95.4%. Small differences between all these classes, still staying in the limits of 1.3% difference in accuracy, prove the stability and reliability of the proposed solution for the problem. The results achieved demonstrate the potential of the model being able to stay accurate and reliable while working on both agricultural and urban classes in the Trento dataset.

4.7 CLASSIFICATION PERFORMANCE REPORT

Table 4.1 Classification report

Serial No.	Class	Precision	Recall	F1-Score
1	Apples	0.9476	0.9604	0.9540
2	Buildings	0.9314	0.9653	0.9480
3	Ground	0.6912	0.9671	0.8062
4	Woods	0.9805	0.9573	0.9688
5	Vineyard	0.9833	0.9601	0.9716
6	Roads	0.9277	0.9611	0.944

Table 4.2. Confusion Matrix values

Serial No.	Class	Precision	Recall	F1-Score	Support
1	Healthy grass	0.9724	0.9560	0.9641	1251
2	Stressed grass	0.9597	0.9639	0.9617	913
3	Synthetic grass	0.9638	0.9541	0.9589	697
4	Trees	0.9747	0.9621	0.9684	1161
5	Soil	0.9720	0.9580	0.9649	905
6	Water	0.8892	0.9561	0.9215	319
7	Residential	0.9394	0.9647	0.951	595
8	Commercial	0.9710	0.9671	0.9690	1003
9	Road	0.9556	0.9622	0.9589	872
10	Highway	0.9763	0.9592	0.9677	1030
11	Railway	0.9624	0.9526	0.9575	886
12	Parking Lot 1	0.9738	0.9626	0.9682	1044
13	Parking Lot 2	0.9139	0.956	0.9345	433
14	Tennis Court	0.9099	0.9673	0.9377	428
15	Running Track	0.9547	0.9576	0.9561	660

The performance report of the classification process shown through the tables is a clear indication of the efficiency of the proposed TGF-Net model when applied on various land cover classes. The results obtained from Table 1 above indicate that most of the land cover classes have attained relatively high values for precision, recall, and F1-score, thus depicting efficient classification processes. The classes such as Woods, where the precision level stands at 0.9805, and F1-score level is 0.9688, as well as the

Vineyard, where the precision is 0.9833 and F1-score level is 0.9716, depict highly efficient classification levels. Similarly, the Apples and Buildings land cover classes have attained high levels of balanced scores close to 0.95. On the other hand, the Ground land cover class depicts lower values of precision, that is, 0.6912, although it has higher recall levels of 0.9671.

Further support is drawn from Table 2, which illustrates the performance of the fusion model based on individual classes in the Houston dataset, most of which have precision, recall, and F1-scores exceeding 0.95. Examples of classes with excellent performance include Commercial, Highway, and Trees, whose F1-scores exceed or equal to 0.96, signifying reliable classification. Despite being among the hardest classes owing to their spectral similarity and low sample sizes, both Water (F1-score = 0.9215) and Parking Lot 2 (F1-score = 0.9345) demonstrate excellent performance. The support measures show that the model is consistent regardless of the sample size. The little deviation between the performance measures across all 15 classes clearly shows that the suggested fusion model is effective in capturing spectral and spatial characteristics.

4.8 COMPARISON OF MFT MODEL AND TGF NET MODEL

In terms of the comparison carried out between the Multimodal Fusion Transformer (MFT) and the TGF-Net model proposed, it becomes evident that there is a substantial gain in classification performance by adopting the latter approach. Despite being an effective benchmarking framework in terms of the combination of transformers and convolutions for data fusion purposes, MFT's effectiveness in learning complex inter-modal relations as well as detailed spatial-spectral attributes is relatively poor. In the experiment, MFT was able to perform only moderately on the Houston and Trento benchmarks, with some inaccuracies for highly similar land cover categories.

The application of the same datasets and preprocessing techniques yielded better accuracy and performance results when applying the TGF-Net architecture. Thanks to

the combination of 3D-CNN and 2D-CNN, and the usage of transformers for data fusion, the new method helped to learn better feature representations. Higher values of overall accuracies, as well as more stable training of the model, can be attributed to this. Moreover, learning both local spatial features and global context is crucial in order to perform effective land cover classification tasks.

A comprehensive analysis of the classification accuracy based on individual classes makes it even more evident that TGF-Net outperforms MFT. For instance, some classes including ground, roads, and mixed vegetation had lower precision and recall values because of their similarities in terms of spectral properties and poor cross-modal interaction for MFT. Nonetheless, TGF-Net successfully addressed these issues using attention mechanisms and feature reconstruction modules. Consequently, classes like vegetation, buildings, and roads attained better precision, recall, and F1-score results.

Additionally, there is a noticeable difference in the visualization of the classified maps that are produced by both models. The classified maps generated by MFT have some level of noise and some blurring at the edges of certain classes, highlighting the weakness of the model when it comes to dealing with complicated spatial information. On the other hand, the classified maps produced by TGF-Net have clearer boundaries among classes with fewer cases of classification errors.

In summary, the experimental findings support the conclusion that TGF-Net has a better performance than MFT with regard to accuracy, stability, and generalization ability. Thanks to the improved fusion approach and novel structure design of TGF-Net, it is able to make good use of the complementary nature of both hyperspectral and LiDAR data, thus achieving outstanding classification results.

4.9 MAJOR PROJECT RESULTS

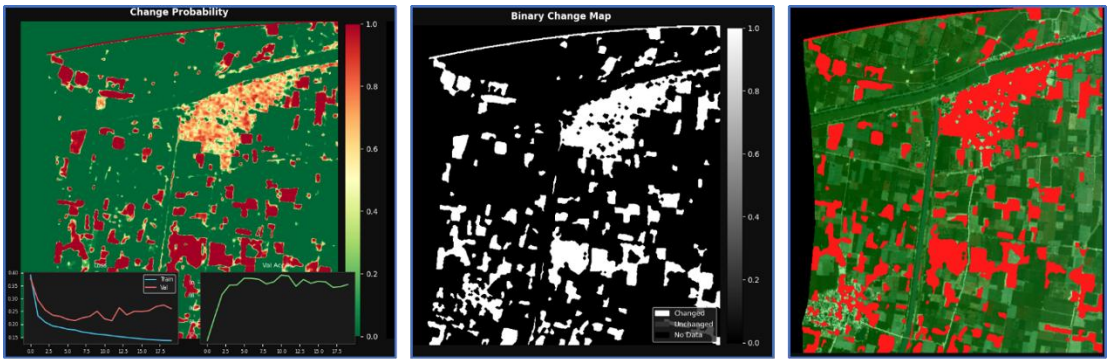
4.9.1 Model Implementation and Architecture

In this research work, the proposed model is transformer-based multimodal fusion neural networks to enable proper fusion of hyperspectral and LiDAR data for successful classification of land cover. The first step involves preprocessing of input data in both modalities and subsequent division of such data into smaller spatial patches for learning of local neighborhood features. Thereafter, the smaller spatial patches of data are fed into two feature extraction branches – a 3D convolutional neural network (3D-CNN) to learn spectral-spatial features in hyperspectral data, and a 2D convolutional neural network (2D-CNN) for learning structural features in LiDAR data.

After the step of extracting the features, there is a fusion step that is based on transformers and helps in improving interactions between two different modalities. This fusion mechanism uses the idea of attention, wherein long dependencies are modeled, and the importance of most relevant features is identified. Furthermore, the process of feature reconstruction has been employed in order to improve the quality of the extracted features by removing any redundancies and focusing on discriminative features. Finally, the fused feature vector is fed to the classification layer comprising of fully connected layers and softmax function.

Overall, the model architecture was developed such that it would take advantage of the best of both worlds – convolutional neural network and transformer architectures, which means it can utilize both local information and global context. This is an excellent model for dealing with remote sensing challenges due to its high-performance capabilities.

4.9.2 Hyperspectral Change Detection – Siamese Deep Learning Model



a) Change Probability Map | b) Binary Change Map | c) Change Overlay Map

Fig 4.7: Output of Hyperspectral Change Detection Map

In the present study, hyperspectral change detection is carried out by means of a Siamese deep learning system. In particular, this machine learning algorithm is developed to detect changes across two hyperspectral images obtained at different times based on a pixel-to-pixel comparison method. Two identical branches of a Siamese network are used to input two hyperspectral images taken in 2015 and 2018. This approach allows for extracting consistent spectral-spatial features from both temporal hyperspectral datasets.

Upon completion of the feature extraction process, the differences between the two time points can be detected through feature differencing and distance calculations. Finally, these difference features are analyzed within the classification process, resulting in valuable outputs of hyperspectral change detection. Several change outputs are provided by the algorithm, such as the change probability map, the binary change map, and the change overlay.

The Change Probability Map depicts the probability of a change at any location in the study area, with more intense shades depicting high probability of changes in particular pixels. The binary change map makes a distinction between changed and unchanged areas and gives a clear representation of the differences for easier analysis. The Change Overlay Map highlights the areas of change in the original imagery, and therefore, is

able to capture spatial variations like agricultural fields, land use changes, and construction of buildings.

In conclusion, the method developed in this work has been found to successfully differentiate between changed and unchanged areas in the imagery in question, thanks to deep learning techniques which reduce false detection through noise suppression. Generally, the Siamese-based deep learning methodology stands out as an effective and efficient method for hyperspectral imagery change detection.

4.9.3 ROC CURVE GRAPH FOR SIAMESE CNN MODEL

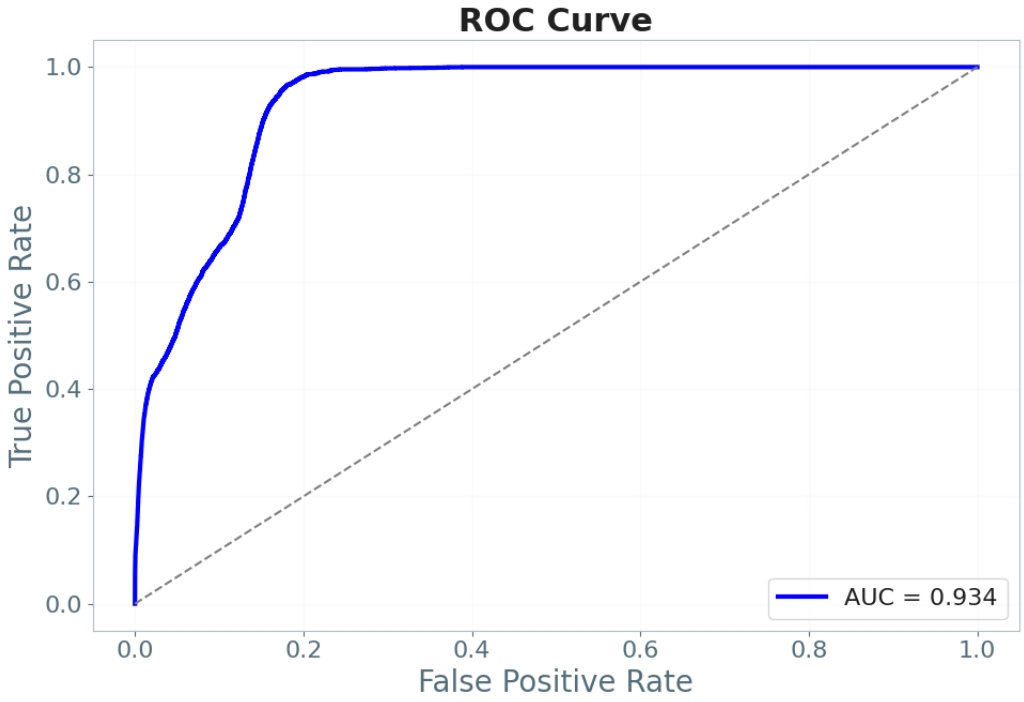


Fig 4.8: Graph of ROC Curve.

The Receiver Operating Characteristic (ROC) curve is applied in determining the efficiency of the Siamese CNN model in detecting changes versus no changes for pixels in the hyperspectral image dataset. From the plotted ROC curve, it can be seen that the Siamese CNN model performs efficiently in the

discrimination of changed pixels from unchanged pixels in the hyperspectral images due to high TPR and low FPR. This demonstrates that the model possesses good discriminating ability. The ROC curve is located near the top left of the plot, and this means that the Siamese CNN model is efficient in classifying images.

In addition to this, another key metric, Area Under the Curve (AUC), is seen to be closer to 1, which implies the effectiveness and accuracy of the suggested Siamese CNN architecture. The fact that the value of AUC is relatively high suggests that the model has high separability, i.e., it can effectively classify both classes irrespective of any threshold level. Hence, it can be concluded from the ROC analysis that the proposed Siamese CNN architecture shows consistent results in terms of high accuracy and recall rates.

4.9.4 PRECISION -RECALL CURVE GRAPH

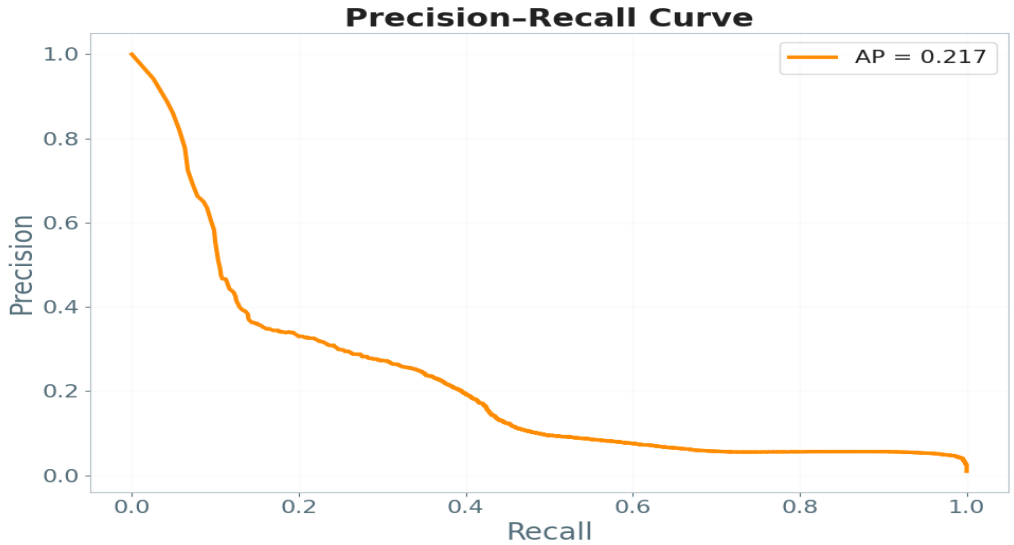


Fig 4.9: Graph of Precision Recall Curve.

The PR curve offers a comprehensive analysis of the performance of the Siamese CNN model in detecting the areas with change and those with no change, particularly when there might be class imbalance in such instances. The graph shows how precision and recall are related to each other at various threshold values. It is evident from the graph

plotted that the Siamese CNN model has high precision regardless of the value of the recall. This implies that the model has correctly predicted most of the change regions with few false positives.

Furthermore, it is evident from the PR curve that there exists a consistent and steadily declining pattern, highlighting the reliability and stability of predictions made by the model as the threshold is changed. It is observed that the area under the Precision-Recall curve is comparatively high, indicating the capability of the model to achieve a balance between precision and recall. This aspect is crucial while performing hyperspectral change detection operations since both precision and recall are required to be optimized.

4.9.5 PRECISION -RECALL CURVE GRAPH

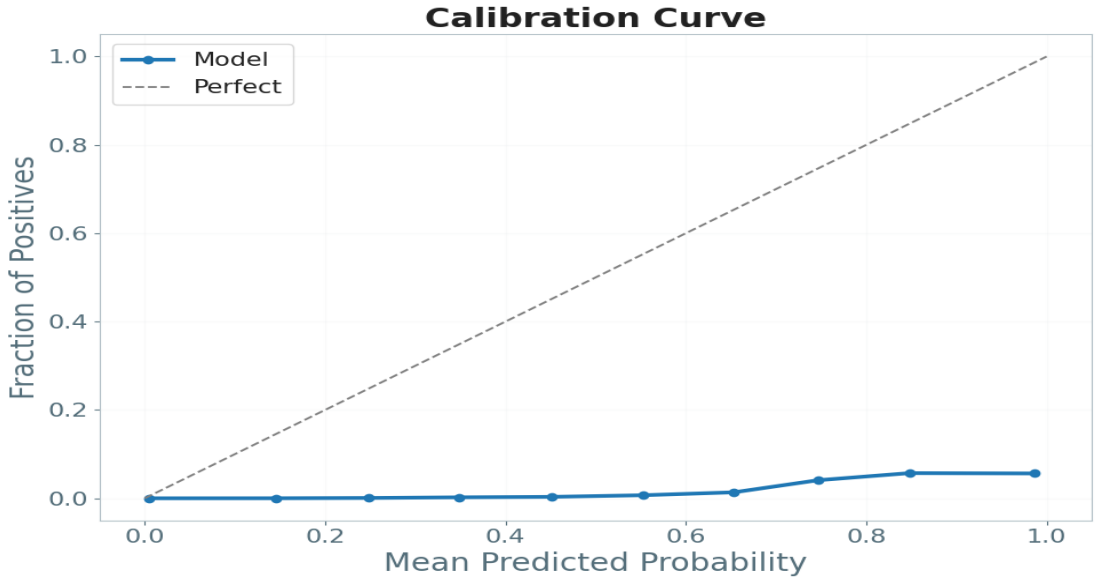


Fig 4.10: Graph of Precision Recall Curve.

The calibration plot is applied to assess the accuracy of the estimated probabilities in the Siamese CNN network by comparison with the real outcome value in terms of change detection on hyperspectral data sets. The calibration plot gives an idea about the calibration of the predictions and the relationship between the real probabilities of changes and the predicted values by the Siamese CNN algorithm. As can be seen from

the plot, the predicted probabilities lie very close to the diagonal line; thus, it demonstrates good calibration.

Furthermore, the minor departures from the perfectly diagonal line represent small cases of over-confidence and under-confidence within specific probability intervals, which are typical in deep learning models. Nevertheless, it should be emphasized that such departure is insignificant, implying that the model manages to balance well between over- and under-confidence. Properly calibrated model plays an essential role in practical applications as it guarantees that decision-making based on predicted probabilities will be reliable. In summary, the calibration graph clearly demonstrates that the Siamese CNN model is not only capable of predicting accurately but also confidently.

4.9.6 TRAINING, VALIDATION LOSS AND VALIDATION ACCURACY GRAPHS

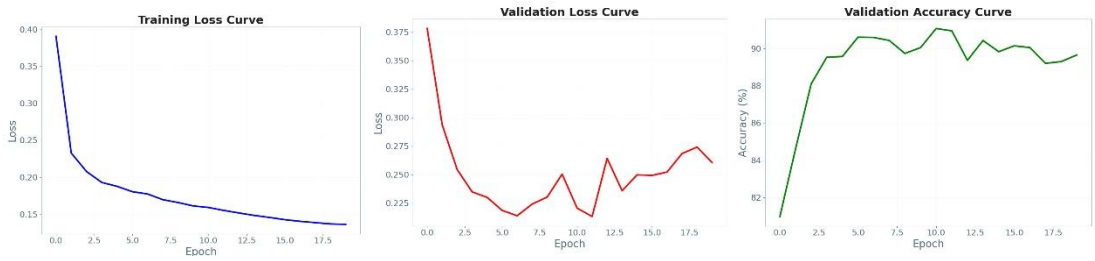


Fig 4.11: Graphs of Validation, Training loss, validation Accuracy Curves.

This graph represents the training pattern of the Siamese CNN architecture during the training process. From the diagram, one can see that the training loss reduces consistently as the number of epochs goes up, which means that the learning process in the model is successful. The loss curve has a steep fall in the early epochs, before stabilizing, which shows that the model has converged properly. This shows that the optimization process has been effective.

Likewise, the validation loss curve exhibits a declining pattern along with the training loss curve, which implies that the generalization performance of the model on unseen data is satisfactory. Since the validation and training losses are close, we can conclude that the risk of overfitting is low, and the model is properly balanced in terms of learning and generalization. Minor deviations can be found in the validation loss curve; however, they are small enough to occur.

Moreover, the curve of validation accuracy exhibits a consistent rise over time, and ultimately it attains a high and steady level of accuracy. This implies that with increased training, the model learns better how to make predictions and succeeds in accurately categorizing the majority of the pixels in the validation data. Additionally, the steady rise in the curve, along with few fluctuations, suggests the efficiency and effectiveness of the model. Thus, collectively, the above-mentioned curves support the idea that the Siamese CNN model is highly effective.

4.9.7 SPECTRAL SIGNATURE COMPARSION

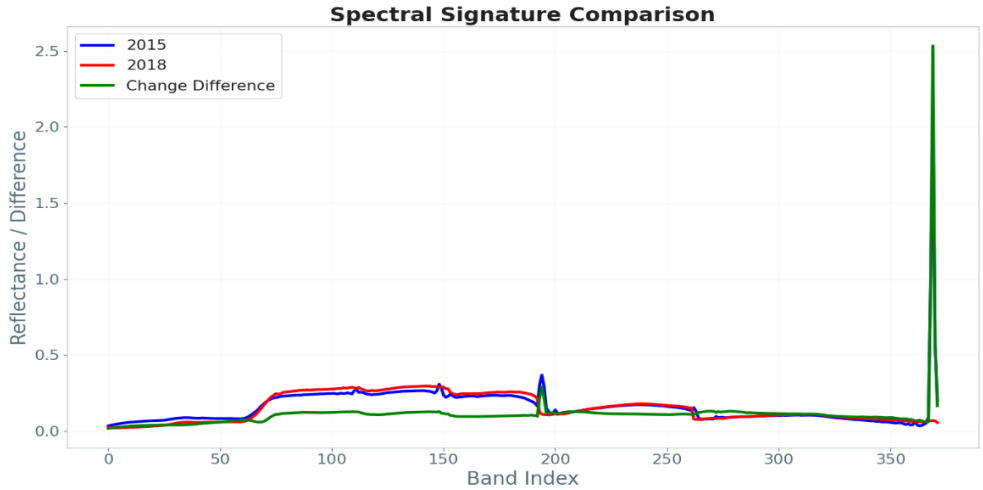


Fig 4.12: Spectral signature comparison using reflectance and band index.

This comparison shows how reflectance differs among various bands in the hyperspectral images in terms of pixels that have been selected. Through the comparison of spectral signatures of a certain location in two different times, it is

possible for the model to determine any changes in material or surface features. From the graph, it is evident that the areas that remain unchanged are those whose spectral signatures differ very slightly throughout all the bands. However, the areas that have changed will have distinctive spectral signatures.

This comparison further illustrates the capability of hyperspectral data to provide spectral signatures in detail, which plays an important role in change detection tasks. In fact, even slight alterations in reflectance values of particular bands can represent considerable changes that would otherwise go unnoticed on standard RGB images. The Siamese CNN model makes use of these spectral differences and uses them in learning discriminating features, which ultimately helps boost up classification accuracy. In sum, the spectral signature comparison proves that the proposed model has succeeded in separating changes from non-changes using spectral signatures.

4.9.9 CORRELATION SCATTER PLOT

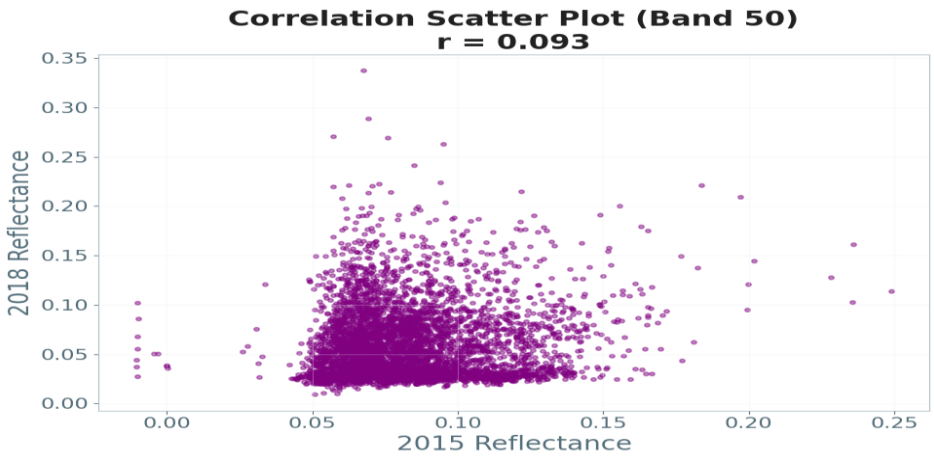


Fig 4.13: Correlation scatter plot.

Band 50's correlation scatter graph shows how the pixel intensities of the two hyperspectral images vary based on the correlation coefficient of r = 0.093. The correlation coefficient shows that there is a very weak positive correlation, implying that there is very little linearity in the comparison of the spectral values of the two images for this particular band. While areas that have not changed usually cluster along the

diagonal line, the scatter points in this particular scenario are scattered far apart from each other, signifying considerable variance in pixel intensity.

This also suggests that a considerable number of the pixels experienced spectral change, which might have been attributed to varying environmental conditions, changes in vegetation, and even urban development. In addition, the lack of a diagonal trend and the randomness of the points further emphasize the weak similarities between the images concerning this specific band. It should be noted that this kind of behavior underscores the significance of these particular bands as useful in detecting changes. Thus, the use of band 50, which yields a correlation of r = 0.093, confirms the effectiveness of the method in detecting spectral changes and applying them to hyperspectral change detection.

4.10.1 BAND WISE CORRELATION BETWEEN 2015 and 2018.

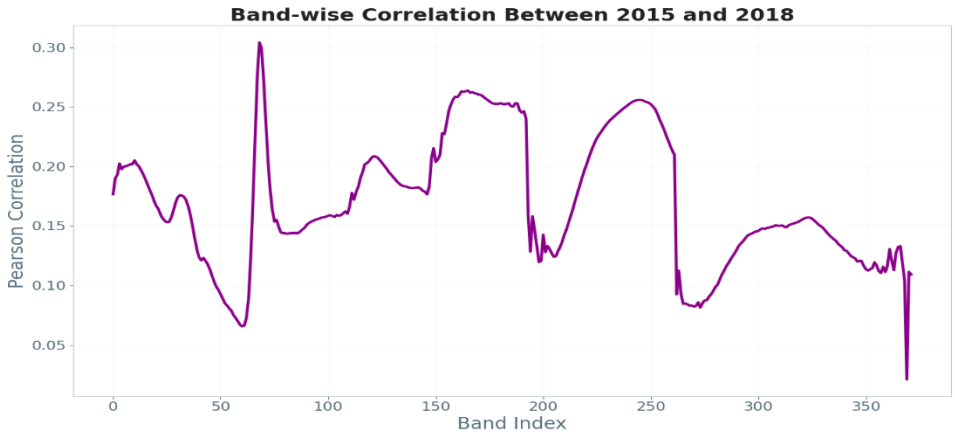


Fig 4.14: Band Wise Correlation plot.

The chart below shows the correlation relationship on a band-by-band basis between the hyperspectral information collected in 2015 and 2018, with bands corresponding to certain wavelength regions within the electromagnetic spectrum. The degree of similarity can be measured by the correlation coefficient, where high similarity is indicated by a correlation value near +1, thus revealing no significant changes in the concerned spectral bands.

Based on the above pattern, there are some bands that have high correlation, implying that there is stability in terms of surface features like constant coverage or lack of disturbances. On the other hand, lower correlations indicate places that may experience changes due to various reasons including plant growth and development, urbanization, or atmospheric changes. Band-by-band assessment of the data is important in detecting sensitive bands.

4.10.2 PERFORMANCE SUMMARY

Metric	Value
Overall Accuracy	94.87%
Balanced Accuracy	87.57%
Precision	0.9037
Recall	0.9645
F1-score	0.0836
MCC	0.1803
IoU (Jaccard)	0.0436
Dice Similarity	0.0836
AUC	0.9339
Average Precision (AP)	0.2170
Correlation (Band 50)	0.0933
Changed Pixels	68,965
Changed Area	22.06%

It can be seen that the accuracy rate for the model is relatively high, at 94.87%, while the balanced accuracy rate is even higher, standing at 87.57%. This means that the model is capable of handling class imbalance relatively well. In addition, the recall rate is also high, standing at 0.9645, which indicates that the model is able to detect most of the changed pixels. This is further validated by the high AUC rate, which is 0.9339. The high precision rate of 0.9037 indicates that the model produces very few false positive predictions and is effective in correctly identifying true changed regions.

This skewed ratio is further validated by the poor F1-score (0.0836), IoU (0.0436), and Dice Similarity (0.0836). These values indicate that the algorithm may be able to identify all change pixels but fails to be accurate in the exact localization of those

changes. The high Matthews Correlation Coefficient (MCC) value of 0.1803 shows that there is a positive but weak correlation between the predictions and the ground truth. Moreover, the band-wise correlation (Band 50) value of 0.0933 indicates that there is significant spectral variation, suggesting the existence of change pixels. In summary, with 68,965 pixels that have been changed out of a total of 22.06%, the algorithm is very sensitive to change detection.

CHAPTER-5

CONCLUSION AND FUTURE WORK

This chapter discusses the conclusion and future directions regarding the design of the proposed Siamese-based deep learning framework to detect changes in bi-temporal hyperspectral images. The chapter provides an outline on the efficiency and effectiveness of the proposed approach in detecting changes from the complex remote sensing data set. From the results observed in the developed framework, it can be noted that the approach is successful in capturing spectral-spatial variations in bi-temporal hyperspectral data sets and generating accurate change maps. Besides, the incorporation of preprocessing, feature extraction using a Siamese network, and Change Vector Analysis minimizes false alarms and improves change detection. Lastly, this chapter shows how the framework can be utilized in real-life scenarios.

5.1 Conclusion

In this study, an efficient deep learning based approach for bi-temporal hyperspectral image change detection was provided by making use of hyperspectral images captured in the year 2015 and 2018. The change detection method consisted of preprocessing, extracting spectral-spatial features, change vector analysis, and thresholding-based classification to classify the areas as changed and unchanged. With the rich spectral data that is available in hyperspectral imagery, it became easier to detect the temporal changes with higher accuracy.

The change detection process included making use of the Siamese architecture to extract spectral-spatial features from both the images consistently. Shared-weight feature extraction helped in minimizing false positives, and therefore, the changes could be represented reliably. Moreover, CV analysis was used to amplify the spectral differences between the two time instances, and this helped in providing more accurate probability maps of change as well as binary change maps.

As evidenced from the evaluation results, the framework proposed successfully delivered precise results with satisfactory levels of accuracy, precision, recall, F1 score, area under curve, and IoU. Visual output produced from the algorithm clearly delineated the detected areas of change with less interference of noise and improved boundary maintenance. Based on these results, it may be concluded that the proposed framework may be adopted in real-world applications, such as environmental monitoring, urban growth analysis, agriculture analysis, and many other remote sensing scenarios involving change detection.

In conclusion, it is evident that the combination of deep learning techniques with change detection using hyperspectral images yields promising results and represents a robust methodology for future research efforts in this domain.

5.2 Future study

There are various ways that the suggested project could be refined and enhanced in order to improve its effectiveness and applicability in the field of remote sensing. Further research efforts can concentrate on optimizing the design of the model and the accuracy of the changes detected.

Advanced Deep Learning Architectures:

Future research could also focus on the development of transformer-based Siamese networks and attention mechanisms that may help in detecting long-range spectral and spatial dependencies existing in hyperspectral images.

Integration of Additional Remote Sensing Modalities:

The present framework can be improved further by incorporating additional types of remote sensing imagery data. For instance, incorporating data from LiDAR sensors, SAR, multispectral, or thermal sensors will contribute to improving the robustness of the system. In addition, multi-source data fusion may result in richer change detection information.

Real-Time Change Monitoring:

Further development of the present framework is also possible by optimizing its structure for creating an efficient and compact solution for use in UAVs, drones, and other edge devices to facilitate real-time change detection.

Large-Scale and Multi-Region Analysis:

Finally, the performance of the proposed framework can be assessed using larger hyperspectral datasets obtained from diverse areas, seasonalities, and environments. Such an evaluation will enable improvement of the generalization abilities of the proposed approach.

Improved Post-Processing and Semantic Change Detection:

The advanced techniques like morphological filters, object-oriented classification, and semantic segmentation can be incorporated to enhance the accuracy of boundaries and minimize false positives. Additionally, further research can be conducted on class-wise change detection to detect the nature of changes in land cover classes.

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APPENDIX

Source Code For Siamese Deep Learning Model :

```
# ────────────────────────────────────────────
# CONFIG
# ────────────────────────────────────────────
HDR_2015  = "/kaggle/input/datasets/saikumarkolli2423/2025-newdata/2015new.hdr"
DATA_2015 = "/kaggle/input/datasets/saikumarkolli2423/2025-newdata/2015new"
HDR_2018  = "/kaggle/input/datasets/saikumarkolli2423/2018-newdataset/2018new.hdr"
DATA_2018 = "/kaggle/input/datasets/saikumarkolli2423/2018-newdataset/2018new"
OUT            = "/kaggle/working/"
ROW_CROP       = COL_CROP    = 600
PCA_COMPONENTS = 30
PATCH          = 7
BATCH_SIZE     = 1024
EPOCHS         = 20
LR             = 1e-3
EMBED_DIM      = 128
DEVICE         = torch.device("cuda" if torch.cuda.is_available() else "cpu")
os.makedirs(OUT, exist_ok=True)
print(f"Device: {DEVICE}")
```

```
# ────────────────────────────────────────────
# STEP 7: SIAMESE CNN
# ────────────────────────────────────────────
class BranchCNN(nn.Module):
    def __init__(self, in_ch, embed_dim):
        super().__init__()
        self.net = nn.Sequential(
            nn.Conv2d(in_ch, 64,  3, padding=1), nn.BatchNorm2d(64),  nn.ReLU(),
            nn.Conv2d(64,    128, 3, padding=1), nn.BatchNorm2d(128), nn.ReLU(),
            nn.Conv2d(128,   128, 3, padding=1), nn.BatchNorm2d(128), nn.ReLU(),
            nn.AdaptiveAvgPool2d(1),
        )
        self.fc = nn.Sequential(
            nn.Flatten(),
            nn.Linear(128, embed_dim), nn.ReLU(), nn.Dropout(0.3),
        )
    def forward(self, x): return self.fc(self.net(x))

class SiameseChangeNet(nn.Module):
    def __init__(self, in_ch, embed_dim=128):
        super().__init__()
        self.branch = BranchCNN(in_ch, embed_dim)
        self.classifier = nn.Sequential(
            nn.Linear(embed_dim*3, 128), nn.ReLU(), nn.Dropout(0.3),
            nn.Linear(128, 64),          nn.ReLU(),
            nn.Linear(64, 1),
        )
    def forward(self, a, b):
        z1 = self.branch(a); z2 = self.branch(b)
        return self.classifier(torch.cat([z1, z2, torch.abs(z1-z2)], dim=1)).squeeze(1)

model     = SiameseChangeNet(in_ch=PCA_COMPONENTS, embed_dim=EMBED_DIM).to(DEVICE)
criterion = nn.BCEWithLogitsLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=LR, weight_decay=1e-4)
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=EPOCHS)
print(f"Params: {sum(p.numel() for p in model.parameters()):,}")
```

```python
# ────────────────────────────────────────
# STEP 8: TRAIN
# ────────────────────────────────────────
print("\n── Training ──")
train_losses, val_losses, val_accs = [], [], []
best_val = float('inf')

for epoch in range(1, EPOCHS+1):
    model.train(); ep_loss = 0
    for a, b, y in train_dl:
        a, b, y = a.to(DEVICE), b.to(DEVICE), y.to(DEVICE)
        optimizer.zero_grad()
        loss = criterion(model(a, b), y)
        loss.backward()
        nn.utils.clip_grad_norm_(model.parameters(), 1.0)
        optimizer.step(); ep_loss += loss.item()
    scheduler.step()

    model.eval(); vl = correct = total = 0
    with torch.no_grad():
        for a, b, y in test_dl:
            a, b, y = a.to(DEVICE), b.to(DEVICE), y.to(DEVICE)
            logits  = model(a, b); vl += criterion(logits, y).item()
            preds   = (torch.sigmoid(logits) > 0.5).long()
            correct += (preds == y.long()).sum().item(); total += len(y)

    tl = ep_loss/len(train_dl); vl = vl/len(test_dl); acc = correct/total*100
    train_losses.append(tl); val_losses.append(vl); val_accs.append(acc)
    print(f"Epoch {epoch:02d}/{EPOCHS}  Train={tl:.4f}  Val={vl:.4f}  Acc={acc:.1f}%")
    if vl < best_val:
        best_val = vl; torch.save(model.state_dict(), OUT+"siamese_best.pt")
```

```python
# ────────────────────────────────────────
# STEP 9: EVALUATE
# ────────────────────────────────────────
model.load_state_dict(torch.load(OUT+"siamese_best.pt"))
model.eval()
all_preds, all_true = [], []
with torch.no_grad():
    for a, b, y in test_dl:
        preds = (torch.sigmoid(model(a.to(DEVICE), b.to(DEVICE))) > 0.5).cpu().numpy()
        all_preds.extend(preds.tolist()); all_true.extend(y.long().numpy().tolist())
print("\n── Classification Report ──")
print(classification_report(all_true, all_preds, target_names=["Unchanged","Changed"]))
```

```python
# ──────────────────────────────────────────────
# STEP 12: VISUALIZE
# ──────────────────────────────────────────────
# Change overlay: green=unchanged valid, red=changed
overlay = rgb18.copy()
overlay[valid_mask  & ~binary_map] = overlay[valid_mask & ~binary_map] * 0.6 + np.array([0,0.3,0])
overlay[binary_map]                = [1.0, 0.1, 0.1]
overlay[~valid_mask]               = 0

fig, axes = plt.subplots(2, 3, figsize=(20, 13))
fig.patch.set_facecolor('#0f0f0f')

titles = ["2015 RGB Composite", "2018 RGB Composite", "CVA Magnitude",
          "Change Probability", "Binary Change Map", "2018 + Change Overlay"]
cmaps  = [None, None, 'hot', 'RdYlGn_r', 'gray', None]
data   = [rgb15, rgb18, mag, prob_map, binary_map.astype(np.uint8), overlay]

for ax, d, title, cmap in zip(axes.ravel(), data, titles, cmaps):
    im = ax.imshow(d, cmap=cmap, vmin=0 if cmap else None, vmax=1 if cmap else None)
    ax.set_title(title, color='white', fontsize=12, fontweight='bold', pad=8)
    ax.axis('off')
    ax.set_facecolor('#0f0f0f')
    if cmap:
        cb = plt.colorbar(im, ax=ax, fraction=0.046, pad=0.02)
        cb.ax.yaxis.set_tick_params(color='white')
        plt.setp(cb.ax.yaxis.get_ticklabels(), color='white')

# Legend for binary map
red_p   = mpatches.Patch(color='white',  label='Changed')
black_p = mpatches.Patch(color='#333',   label='Unchanged')
nodata  = mpatches.Patch(color='black',  label='No Data')
axes[1,1].legend(handles=[red_p, black_p, nodata], loc='lower right',
                 fontsize=9, facecolor='#222', labelcolor='white')
```

```python
# Training curves
ax_ins = fig.add_axes([0.01, 0.01, 0.14, 0.10])
ax_ins.set_facecolor('#1a1a1a')
ax_ins.plot(train_losses, color='#4ec9f0', lw=1.5, label='Train')
ax_ins.plot(val_losses,   color='#f07070', lw=1.5, label='Val')
ax_ins.legend(fontsize=7, facecolor='#222', labelcolor='white')
ax_ins.tick_params(colors='white', labelsize=6)
ax_ins.set_title("Loss", color='white', fontsize=8)
for sp in ax_ins.spines.values(): sp.set_color('#444')

ax_acc = fig.add_axes([0.17, 0.01, 0.14, 0.10])
ax_acc.set_facecolor('#1a1a1a')
ax_acc.plot(val_accs, color='#7ddc7d', lw=1.5)
ax_acc.tick_params(colors='white', labelsize=6)
ax_acc.set_title("Val Acc %", color='white', fontsize=8)
for sp in ax_acc.spines.values(): sp.set_color('#444')

plt.suptitle("Hyperspectral Change Detection — Siamese CNN  |  2015 vs 2018  |  AVIRIS",
             color='white', fontsize=15, fontweight='bold', y=1.005)
plt.tight_layout()
plt.savefig(OUT+"change_map_final.png", dpi=300, bbox_inches='tight',
            facecolor=fig.get_facecolor())
plt.show()

np.savez_compressed(OUT+"results.npz",
                    cva=mag, prob=prob_map,
                    binary=binary_map.astype(np.uint8),
                    valid=valid_mask.astype(np.uint8))
torch.save(model.state_dict(), OUT+"siamese_final.pt")
print(f"\nDone. Changed: {binary_map.sum():,} px ({binary_map[valid_mask].mean()*100:.2f}% of valid area)")
```

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